



# **basic education**

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Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

## **Curriculum and Assessment Policy Statement: Technical Occupational Year 2 -4**

**LIFE SKILLS:  
NATURAL SCIENCES**

PUBLIC COMMENT

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## **SECTION 1:**

### **INTRODUCTION TO THE CURRICULUM AND ASSESSMENT POLICY STATEMENT: TECHNICAL OCCUPATIONAL**

- 1.1 Background
- 1.2 Overview
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## SECTION 2:

### INTRODUCTION TO NATURAL SCIENCES

Science as we know it today has roots in African, Arabic, Asian, European and American cultures. It has been shaped by the search to understand the natural world through observation, testing and providing of ideas, and has evolved to become part of the cultural heritage of all nations. In all cultures and in all times people have wanted to understand how the physical world works and have needed explanations to satisfy them.

#### 2.1 What is Natural Sciences?

Natural Sciences is part of a selection of study areas that make up the Life Skills Learning Programme as specified for the GCE TO qualification. Science is a systematic way of doing investigations through applying scientific knowledge to find explanations for phenomena. It is the **understanding of how things work in the world around us** and also support the learners to **understand their responsibility towards themselves, others and the environment**. It has a **direct link** with the **knowledge and skills taught in** various vocational subjects. When teaching Natural Sciences teachers must use the relevant occupational skills workshops as a context when addressing science content or scientific investigation skills. **The content in Natural Sciences is organised under the follow four knowledge strands; Life** and Living, Matter and Material, Energy and Changes, and Planet Earth and Beyond

|                              |                                                                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal</b>                  | Pursuit of new knowledge and understanding of the world around us and of natural phenomena                                                                                                                                 |
| <b>Focus</b>                 | Focus is on understanding the natural world                                                                                                                                                                                |
| <b>Developmental methods</b> | Discovery through carrying out investigations                                                                                                                                                                              |
| <b>Major processes</b>       | Investigative and logical processes <ul style="list-style-type: none"><li>• planning investigations</li><li>• conducting investigations and collecting data</li><li>• evaluating data and communicating findings</li></ul> |
| <b>Evaluation methods</b>    | Analysis, generalisation and creation of theories                                                                                                                                                                          |

Natural Sciences is not offered as a subject in year 1.

## 2.2 Topics to be studied in Natural Sciences (Natural Sciences is NOT offered in year 1.)

|                                | Year 2                       |               | Year 3                             |               | Year 4                                     |               |
|--------------------------------|------------------------------|---------------|------------------------------------|---------------|--------------------------------------------|---------------|
| TERM 1                         | TOPIC                        | TIME IN WEEKS | TOPIC                              | TIME IN WEEKS | TOPIC                                      | TIME IN WEEKS |
| <b>Life and living</b>         | Living and non-living things | 3 weeks       | Digestive system                   | 3 weeks       | Sense Organs                               | 2weeks        |
|                                | Structure of plants          | 2 weeks       | Nutrients in food                  | 2weeks        | Human skeleton                             | 2weeks        |
|                                | Structure of animals         | 2 weeks       | Nutrition                          | 1 week        | Human Reproductive System                  | 4weeks        |
|                                | What plants need to grow     | 1 week        | Food processing                    | 2 weeks       |                                            |               |
| <b>Assessment</b>              |                              | 2 weeks       |                                    | 2 weeks       |                                            | 2 weeks       |
| <b>TERM 2</b>                  |                              |               |                                    |               |                                            |               |
| <b>Mater and Materials</b>     | Phases of matter             | 3 weeks       | Mixtures                           | 2 weeks       | Acids and bases                            | 3weeks        |
|                                | Metals and non-metals        | 3 weeks       | Solutions as special mixtures      | 2 weeks       | Process to purify water                    | 3weeks        |
|                                | Processing material          | 2 weeks       | Dissolving                         | 2 weeks       | Impact of weather on matter and materials. | 2weeks        |
|                                |                              |               | The water cycle                    | 2 weeks       |                                            |               |
| <b>Assessment</b>              |                              | 2 weeks       |                                    | 2 weeks       |                                            | 2 weeks       |
| <b>TERM 3</b>                  |                              |               |                                    |               |                                            |               |
| <b>Energy and Change</b>       | Energy                       | 2 weeks       | Energy and Temperature             | 2 weeks       | Electric circuits                          | 2 weeks       |
|                                | Energy Sources               | 2 weeks       | Stored energy in fuels             | 1 week        | Systems to solve problems                  | 2 weeks       |
|                                | Energy Transfer              | 2 weeks       | Burning Fuels and safety with fire | 2weeks        | Mains electricity                          | 2 weeks       |
|                                | Sound and Noise pollution    | 1 week        | Energy and Electricity             | 2 weeks       | Saving Cost of Generating Electricity      | 1 week        |
| <b>Assessment</b>              |                              | 2weeks        |                                    | 2 weeks       |                                            | 2 weeks       |
| <b>TERM 4</b>                  |                              |               |                                    |               |                                            |               |
| <b>Planet Earth and Beyond</b> | Planet Earth                 | 4 weeks       | Movements of the Earth and Planets | 5 weeks       | Revision                                   | 4 weeks       |
|                                | The Sun                      | 2 weeks       | Systems for looking into Space     | 2 weeks       |                                            |               |
|                                | The Moon                     | 1 week        |                                    |               |                                            |               |
| <b>Assessment</b>              |                              | 2 weeks       |                                    | 2 weeks       |                                            |               |

### 2.3 Specific Aims:

There are **three** Specific Aims in Natural Sciences.

#### **Specific Aim 1: 'Doing Science'**

Learners should be able to complete investigations, analyse problems and use practical processes and skills in designing and evaluating solutions.

#### **Specific Aim 2: 'Understanding and Connecting Ideas'**

Learners should have a grasp of scientific, technological and environmental knowledge and be able to apply it in new contexts.

#### **Specific Aim 3: 'Science, Technology and Society'**

Learners should understand practical uses of Natural Sciences in society and the environment and have values that make them caring and creative citizens.

#### **Major Process Skills to be taught in Natural Sciences**

- Accessing and recalling information – being able to use a variety of sources to acquire information, and to remember relevant facts and key ideas, and to build a conceptual framework
- Observing – noting in detail objects, organisms and events
- Comparing – noting similarities and differences between things
- Measuring – using measuring instruments such as rulers, thermometers, clocks and syringes (for volume)
- Sorting and classifying – applying criteria in order to sort items into a table, mind-map, key, list or other format
- Raising questions – being able to think of, and articulate relevant questions about problems, issues, and natural phenomena
- Predicting – stating, before an investigation, what you think the results will be for that particular investigation
- Hypothesizing – putting forward a suggestion or possible explanation to account for certain facts. A hypothesis is used as a basis for further investigation which will prove or disprove the hypothesis
- Planning investigations – thinking through the method for an activity or investigation in advance. Identifying the need to make an investigation a fair test by keeping some things (variables) the same whilst other things will vary
- Doing investigations – this involves carrying out methods using appropriate apparatus and equipment, and collecting data by observing and comparing, measuring and estimating, sequencing, or sorting and classifying. Sometimes an investigation has to be repeated to verify the results.

- Recording information – recording data from an investigation in a systematic way, including drawings,
- Descriptions, tables and graphs
- Interpreting information – explaining what the results of an activity or investigation mean (this includes reading skills)

## 2.4 Requirements for Natural Sciences as a subject

### 2.4.1 Time Allocation

The total number of hours allocated for Natural Sciences as part of the Life Skills Learning Programme is 1 hour and 30 minutes in a five-day cycle.

The table below provides the time allocated and mark weighting for the different study areas.

| Study Areas             | Time allocation per week     | Time allocation per year and % weighting |                         |                         |                         |
|-------------------------|------------------------------|------------------------------------------|-------------------------|-------------------------|-------------------------|
|                         |                              | Year 1                                   | Year 2                  | Year 3                  | Year 4                  |
| Personal Well-being     | 2½ hours                     | 100 hours<br>60%                         | 100 hours<br>50%        | 100 hours<br>50%        | 75 hours<br>50%         |
| Physical Education      | 1 hour                       | 40 hours<br>20%                          | 40 hours<br>15%         | 40 hours<br>15%         | 30 hours<br>15%         |
| Creative Arts           | 1 hour                       | 40 hours<br>20%                          | 40 hours<br>15%         | 40 hours<br>15%         | 30 hours<br>15%         |
| <b>Natural Sciences</b> | <b>1½ hours</b>              | <b>x</b>                                 | <b>60 hours<br/>20%</b> | <b>60 hours<br/>20%</b> | <b>45 hours<br/>20%</b> |
| Total                   | 4 ½ hours Y1<br>6 hours Y2-4 | 180 hours                                | 240 hours               | 240 hours               | 180 hours               |
| Weeks                   |                              | 40                                       | 40                      | 40                      | 40                      |



## 2.4.2 Resources

Resources needed for teaching Natural Sciences are listed against each topic in order to assist teachers with planning and preparation. The list is a guide and suitable alternative tools and materials may be used. Every learner must have his/her own textbook. Teachers should ensure that a system is in place for recovering textbooks at the end of every year. Schools must provide secure storage space where textbooks and other equipment can be stored safely.

Ideally every learner should have access to sufficient workplace and equipment to carry out investigations. All safety measurements must be in place in the classrooms when doing investigations and experiments with learners.

With regard to equipment, schools must make every effort to ensure that the essential equipment is provided. Tools, apparatus, material, and consumables must be acquired through a planned budgeting process.

While it is acknowledged that it is not ideal to have to improvise when using equipment, teachers should remember that it is more important for learners to have the experience of carrying out a variety of investigations than to depend on the availability of equipment. In instances where equipment is limited, teachers should be encouraged to improvise. The same skills can be developed using improvised equipment.

The Natural Sciences classroom should be equipped with charts, hand lenses, bio-viewers and relevant bio-strips, reference books, scissors, models, field guides, Identification keys, glass beakers, and if all possible access to appropriate DVDs, DVD player, Internet and data projector.

A selection of simple Science equipment such as glass beakers, test tubes, spirit or bunsen burners, thermometers should also be acquired. Consumable chemicals for practical investigations and experiments should be replenished when necessary.

Fresh plant material can be obtained from the surroundings and teachers should ensure that appropriate plants (e.g. Impatiens) are planted on the school grounds. Teachers must familiarise themselves with the subject content and how to use equipment.

### **2.4.3 Infrastructure, equipment and finances**

Schools must ensure that teachers have the necessary infra-structure, equipment and financial resources for quality teaching and learning.

#### **Infrastructure**

Each class must have a basin and water supply available. The classroom should provide an opportunity for conducting experiments and group work.

#### **Finances:**

Budget and inventory

A budget must be allocated for the subject. The amount will be determined by the number of learners taking the subject across all the years and the nature of the practical work required as stipulated in the curriculum. The budget needs to be revised annually and must consider all resources needed per year. The funding must make provision for maintenance of equipment and the replacement over the years.

A stock inventory must be maintained by the teacher and verified annually by a Senior Management Team member.

### **2.5 Career opportunities**

Teachers must highlight the links between Natural Sciences as a subject and other subjects like Life Skills and all the Occupational subjects offered at the school.

### SECTION 3:

#### OVERVIEW OF TOPICS PER TERM AND ANNUAL TEACHING PLANS

##### 3.1 Content Overview

| STRANDS         | Year 2                                                                                                                                                                                                                                                                                                                                                             | Year 3                                                                                                                                                                                                                                    | Year 4                                                                                                                                                                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Life and living | <b><u>Living and non-living things</u></b><br><b>Living things</b> <ul style="list-style-type: none"> <li>Identify and explain</li> <li>Briefly explain the seven life processes</li> </ul> <b>Non-living</b> <ul style="list-style-type: none"> <li>Identify</li> <li>Explain why things are non-living</li> </ul> <b>Practical: Living and Non-living things</b> | <b><u>Digestive system</u></b> <ul style="list-style-type: none"> <li>Explain concepts</li> <li>Describe structure</li> <li>Name the parts and the organs</li> <li>Describe the features and functions of the parts and organs</li> </ul> | <b><u>Sense Organs</u></b><br><b>Senses</b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Name and describe animal senses adapted for survival</li> </ul>                                                                      |
|                 | <b><u>Structure of plants</u></b> <ul style="list-style-type: none"> <li>Describe the basic structure of plants</li> <li>Identify differences between plants</li> </ul>                                                                                                                                                                                            | <b><u>Nutrients in food</u></b><br><b>Food Groups</b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Categorise food</li> <li>Describe the functions of food groups</li> </ul>                                   | <b><u>Human skeleton</u></b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Function of bones</li> </ul>                                                                                                                        |
|                 | <b><u>Structure of animals</u></b> <ul style="list-style-type: none"> <li>Describe the basic structure of animals</li> <li>Identify differences between animals</li> </ul>                                                                                                                                                                                         | <b><u>Nutrition</u></b> <ul style="list-style-type: none"> <li>Explain a balanced diet</li> </ul>                                                                                                                                         | <b><u>Human Reproductive System</u></b><br><b>Reproduction</b> <ul style="list-style-type: none"> <li>Introduce and explain concepts with examples</li> <li>Explain the physical changes</li> <li>Explain the features in the human organisms</li> </ul> |
|                 | <b><u>What plants need to grow</u></b> <ul style="list-style-type: none"> <li>Explain requirements to growth</li> <li>Explain how plants grow from cuttings and seeds</li> </ul>                                                                                                                                                                                   | <b><u>Food processing</u></b> <ul style="list-style-type: none"> <li>Explain concepts, importance of preserving, types of products, methods and benefits</li> </ul>                                                                       |                                                                                                                                                                                                                                                          |

|                             |                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                              |
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| <b>Matter and Materials</b> | <b><u>Phases of matter</u></b><br><b>Solids, liquids and gases</b> <ul style="list-style-type: none"> <li>Briefly define and describe three phases</li> <li>Distinguish between phases</li> <li>Describe the properties</li> </ul>                                                                         | <b><u>Mixtures</u></b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Distinguish between pure substances and mixtures</li> <li>Separating of mixtures (demonstrate)</li> </ul>                              | <b><u>Acids and bases</u></b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Describe difference between acids and bases</li> <li>Test various household substances with blue and red litmus paper and sort them into acids bases and neutrals</li> </ul> <b>Practical activity</b> |
|                             | <b><u>Metals and non-metals</u></b> <ul style="list-style-type: none"> <li>Define concepts</li> <li>Distinguish between metals and non-metals</li> <li>Properties of metals and non-metals</li> <li>Uses of metals and non-metals</li> </ul>                                                               | <b><u>Solutions as special mixtures</u></b> <ul style="list-style-type: none"> <li>Ingredients of a mixture</li> <li>Explain the difference between a mixture and solution</li> <li>Describe the properties of a mixture</li> </ul>   | <b><u>Process to purify water</u></b><br><b>Clean Water</b> <ul style="list-style-type: none"> <li>Introduce and explain concepts with examples</li> <li>Importance of drinking clean water</li> <li>Methods of cleaning water</li> </ul> <b>Practical Investigation</b>                                     |
|                             | <b><u>Processing material</u></b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Explain processes to make new materials or products</li> <li>List the properties of raw materials used to make new products</li> </ul> <b>Practical: Demonstrate the changes of raw material</b> | <b><u>Dissolving</u></b> <ul style="list-style-type: none"> <li>Explain concept</li> <li>Rate of dissolving</li> </ul> <b>Practical investigation: Rate of dissolving</b>                                                             | <b><u>Impact of weather, temperature and wind on matter and materials.</u></b> <ul style="list-style-type: none"> <li>Impact of weather on materials</li> <li>Ways to protect materials</li> </ul>                                                                                                           |
|                             |                                                                                                                                                                                                                                                                                                            | <b><u>The water cycle</u></b> <ul style="list-style-type: none"> <li>Briefly define concepts</li> <li>Describe the stages of the water cycle</li> <li>Draw the water cycle and describe in your own words</li> </ul> <b>Practical</b> |                                                                                                                                                                                                                                                                                                              |

|                          |                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Energy and Change</b> | <b><u>Energy</u></b> <ul style="list-style-type: none"> <li>• Explain concepts with examples</li> <li>• Different forms/types of energy</li> </ul>                                                                                                   | <b><u>Energy and Temperature</u></b> <ul style="list-style-type: none"> <li>• Explain the concepts with examples</li> <li>• Describe how to read and use a thermometer</li> <li>• Describe how heating and cooling can change materials</li> </ul>                                                                 | <b><u>Electric circuits</u></b> <ul style="list-style-type: none"> <li>• Define concepts</li> <li>• Explain when a circuit is complete</li> <li>• Conductors and Insulators</li> </ul>                                                                 |
|                          | <b><u>Energy Sources</u></b> <ul style="list-style-type: none"> <li>• Explain concepts with examples</li> <li>• Categorise sources of energy into Renewable and Non-Renewable sources</li> <li>• Name the different energy sources</li> </ul>        | <b><u>Stored energy in fuels</u></b> <ul style="list-style-type: none"> <li>• Explain concepts with examples</li> <li>• Explain how energy is stored in food</li> <li>• Discuss the Renewable and Non-renewable sources</li> </ul>                                                                                 | <b><u>Systems to solve problems</u></b> <ul style="list-style-type: none"> <li>• Identify different circuits</li> <li>• Make simple electrical systems</li> </ul>                                                                                      |
|                          | <b><u>Energy Transfer</u></b> <ul style="list-style-type: none"> <li>• Explain where the energy from food comes from</li> <li>• Describe how energy is transferred in a food chain</li> <li>• Identify different types of energy transfer</li> </ul> | <b><u>Burning Fuels</u></b> <ul style="list-style-type: none"> <li>• Investigate, compare &amp; describe how different fuels burn: Example candle</li> </ul> <b><u>Safety with fire</u></b> <ul style="list-style-type: none"> <li>• Identify sources</li> <li>• Identify safety rules when using fuels</li> </ul> | <b><u>Mains electricity</u></b> <ul style="list-style-type: none"> <li>• Describe concepts with examples</li> <li>• Compare the amount of energy released by different energy sources</li> </ul>                                                       |
|                          | <b><u>Sound and Noise pollution</u></b> <ul style="list-style-type: none"> <li>• Explain the concept sound with different examples</li> <li>• Explain sound as noise pollution and the impact on the environment</li> </ul>                          | <b><u>Energy and Electricity</u></b> <ul style="list-style-type: none"> <li>• Briefly define concepts</li> <li>• List the different types of batteries and its uses</li> </ul>                                                                                                                                     | <b><u>Saving Cost of Generating Electricity</u></b> <ul style="list-style-type: none"> <li>• Explain concepts with examples</li> <li>• Describe energy sources and usage in South Africa</li> <li>• Compare different electrical appliances</li> </ul> |

|                               |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                    |                         |
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| Planet Earth<br>and<br>Beyond | <u>Planet Earth</u> <ul style="list-style-type: none"><li>• Explain concepts with examples</li><li>• Describe the features of the Earth</li><li>• Describe the Earth as a planet in space</li></ul> | <u>Movements of the Earth and Planets</u> <ul style="list-style-type: none"><li>• Explain concepts with examples</li><li>• Day and Night rotation</li><li>• The Moon</li></ul>                                                                                     | Revision of term 1 to 3 |
|                               | <u>The Sun</u> <ul style="list-style-type: none"><li>• Describe the unique features of the Sun</li><li>• Describe and compare the size of the Sun to that of the Earth</li></ul>                    | <u>Systems for looking into Space</u> <ul style="list-style-type: none"><li>• Explain how people travel into space</li><li>• Describe how telescopes are used to look into space</li><li>• Describe the largest telescope build and used by South Africa</li></ul> |                         |
|                               | <u>The Moon</u> <ul style="list-style-type: none"><li>• Describe the unique features of the moon</li></ul>                                                                                          |                                                                                                                                                                                                                                                                    |                         |

### 3.2 Content outline per term

Year 2 Term 1

| WEEK  | TOPIC                                                     | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - 3 | Living and Non-living things that share the world with us | <p><b>Living things</b></p> <ul style="list-style-type: none"> <li>Identify and explain different kinds of living things</li> <li>Explain all seven life processes feeding, growing, reproducing, breathing, excreting, sensing, moving, carried out by living plants and animals</li> <li>Explain with examples why some things appear to be dead but will come alive, given the right conditions, e.g. dried yeast, and dried beans, an incubated fertilised egg. Explain why some things were living and are now dead: e.g. dead wood/leaves.</li> </ul> | <p><b>Note to the teacher:</b></p> <p><b>These are only suggested activities, resources and process notes:</b></p> <p><b>Step 1:</b></p> <p>Identify and explain the seven (7) life processes.</p> <ul style="list-style-type: none"> <li>Teacher holds up two objects, one living and the other non-living (e.g. a stone, an animal/plant or pictures). Ask the learners to compare living and non-living organisms</li> <li>Teacher could make notes of learner's ideas and contributions.</li> <li>These are basic definitions of the following concepts:</li> <li>Non-living: Do not grow, do not move and they do not reproduce.</li> <li>Living: Things that are alive and active. They move grow and respond to stimuli</li> <li>Feeding: Food is used to provide energy for living organisms. Green plants make their own food using sunlight. People and animals eat plants and/or other animals.</li> <li>Growing: Babies grow into adults and seedlings grow into plants.</li> <li>Reproducing: Animals and people have babies while new plants grow from seeds.</li> <li>Breathing: Plants, animals and people use the oxygen in the air to turn food into energy.</li> </ul> |

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|  |  | <p><b>Non-living things</b></p> <ul style="list-style-type: none"> <li>Identify and explain different kinds of non-living things:</li> <li>Explain why some things were living and are now dead: e.g. dead wood, dry leaves.</li> </ul> <p><b>Practical activity:</b> Identify and explain the seven life processes of any living organism.</p> | <ul style="list-style-type: none"> <li>Excreting: Waste substances must be removed from the body. Plants, animals and people need to get rid of waste gases and water.</li> </ul> <p>Sensing: All living things respond to changes or notice changes in their surroundings and react to them.</p> <p>Movement: Living things like animals and people move their whole bodies to get from one place to another, plants turn towards the light and their roots grow down into the soil.</p> <p><b>Step 2:</b></p> <p>Learners describe (write short notes on the board) the life processes of living organisms.</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Teacher ensures that all the concepts of the seven life processes are explained and understood.</li> <li><input type="checkbox"/> There are excellent visual resources available on You Tube.</li> </ul> <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>Learners go outside onto school grounds to write down a list of living and non-living things.</li> <li>In their groups they check each other's lists (peer assessment) and discuss. The learners could formulate a definition for living and non-living things. (If needed give guidance)</li> </ul> <p><b>Step 4:</b></p> <ul style="list-style-type: none"> <li>Learners are guided to germinate seeds providing warmth and moisture. It is advised to start germinating seeds during week 1 include this as a observation of the seven life processes.</li> <li>Growing yeast (in warm water with sugar)</li> </ul> |
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|  |  |  | <p><b>Step 5:</b></p> <ul style="list-style-type: none"> <li>• Learners are guided to look at pictures of hatched eggs and discuss conditions of hatching (warmth)</li> <li>• In closing: The teacher describes the life processes that all living organisms have.</li> </ul> <p>Practical activity: This is just a suggestion</p> <p>Choose any living organism and make a poster to illustrate the seven life processes.</p> <p>Resources:</p> <p>Examples and pictures of living and non-living things, including plants, animals, bread, mould, seeds, water, cotton wool, bucket, soil, yeast, pictures of hatched eggs.</p> |
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| 4 -5 | Structure of plants | <p><b>Features of plants</b></p> <ul style="list-style-type: none"> <li>• Explain the following concepts with examples: roots, stems, leaves, flowers, fruits, seeds.</li> <li>• Describe the basic structure of plants: roots, stems, leaves, flowers, fruits, seeds</li> <li>• Identify visible differences and similarities between plants: e.g. small, large, grow in water or soil, produce fruits, flowers; thick/thin/large/small leaves etc.</li> </ul> <p><b>Practical demonstration:</b> Learners must touch, smell and observe the differences in plant structures.</p> | <p>These are only suggested activities, resources and process notes:</p> <p>These are basic definitions of the following concepts:</p> <ul style="list-style-type: none"> <li>• <b>Roots:</b> It is the part of the plant that is usually hidden underground. They hold the plant in the ground and keep it upright. It absorbs water and food from the soil.</li> <li>• <b>Stems:</b> The part of the plant that bears branches, leaves, flowers and fruits. It transports water and minerals from the roots to the leaves.</li> <li>• <b>Leaves:</b> Provide trees with all their food because they turn sunlight into food energy. Leaves also make oxygen in the air that we breath.</li> <li>• <b>Flowers:</b> Flowers are usually the beautiful part of the plant. The colour and fragrance attract insects and bird which play an important role in the reproductive process.</li> <li>• <b>Fruits:</b> Are the fleshy structure of certain plants that may be edible in the raw state for example apples.</li> </ul> <p><b>Seeds:</b> Plants grow from seeds when the conditions are favourable. Seeds are small grains capable of producing a new plan</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Bring a variety of plants and pictures of plants to class.</li> <li>• Learners should identify similarities and differences.</li> <li>• Discuss their findings.</li> <li>• Teacher choose two different plants, draw and label each plant and then compare things that are similar and things that are different between these plants.</li> <li>• Make a table of comparison between the similarities and differences of these two plants.</li> </ul> |
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|  |  |  | <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>• <b>Practical demonstration:</b> Learners must touch, smell and observe the differences in plant structures.</li> </ul> <p><b>Resources:</b></p> <p>Pictures of plants, Examples of a variety of plants</p> |
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| 6 - 7 | Structure of animals | <p><b>Structure of Animals</b></p> <p>Explain the following concepts with examples: Head, Tail, Body, Limbs, Sense organs, Body covering, Vertebrate, Invertebrate</p> <p><b>Features of animals</b></p> <ul style="list-style-type: none"> <li>Describe the basic structure of animals: head, tail, body, limbs, sense organs, body covering, and vertebrate, invertebrate.</li> <li>Identify visible differences between vertebrate and invertebrate animals.</li> </ul> | <p><b>Step 2:</b></p> <p><b>Note to the teacher:</b></p> <p>These are only suggested activities, resources and process notes.</p> <p>These are basic definitions of the following concepts:</p> <ul style="list-style-type: none"> <li><b>Head:</b> It is the part of the animal that has a brain, sensory organs and feeding structures.</li> <li><b>Tail:</b> Most animal have a tail at the back of their body and have different functions for different animals for example monkeys, kangaroos, crocodiles, beavers, fish etc.</li> <li><b>Body:</b> All animals look different for example long legs, short stubby claws, big eyes, thousands of tiny eyes together in one big eye. Animals come in all shapes and sizes.</li> <li><b>Limbs:</b> Animals can walk, run climb or swim using their limbs. Animals can have wings, webbed feet, tentacles, fins, legs, arms, flippers and long slippery bodies.</li> <li><b>Sense organs:</b> Some animals have senses that are much better developed than humans for example excellent smelling, sharp eyes, excellent hearing, echolocation etc.</li> <li><b>Body covering:</b> Different animals need to cover their bodies in different ways. It protects the animals' organs, bones and muscles from the environment. It also helps animals to blend into their environment (camouflage).</li> <li><b>Vertebrate:</b> Animals with a backbone.</li> <li><b>Invertebrate:</b> Animals without a backbone.</li> </ul> |
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|   |                          |                                                                                                                                                                                                                                                                                                                                                                         | <p><b>Step 1</b></p> <ul style="list-style-type: none"> <li>Bring a variety of animals' pictures to class/make use of visual resources for example You Tube.</li> </ul> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>Learners should be able to label the parts of at least one vertebrate and one invertebrate animal.</li> <li>Learners should describe the visible differences between at least three animals applicable to their environment.</li> </ul> <p><b>Resources:</b></p> <p>Visual resources for example pictures, internet etc.</p> |
| 8 | What plants need to grow | <p><b>Requirements for growth of plants</b></p> <ul style="list-style-type: none"> <li>Explain the requirements needed by plants to grow light, warmth, water, gases and fertile soil.</li> <li>Explain and demonstrate how new plants can be grown from cuttings or seeds.</li> </ul> <p><b>Practical Investigation:</b> Growth of plants from seeds and cuttings.</p> | <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li><b>Practical Investigation</b> The growth of plants from seeds and cuttings. Observe measure and record the growth over time and have a discussion over the findings. (<b>Guideline:</b> Plant the seeds or cuttings during week 2 and link it to the seven life processes)</li> </ul> <p><b>Resources:</b></p> <p>Seeds and cuttings, ruler and/or measuring tape, visual media.</p>                                                                                                                |

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| 9 -10                                                                                                                               | The weeks allocated for formal assessment are integrated across the term. |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"> <li>• Practical: 50 %</li> <li>• Test (Pen and Paper): 50 %</li> </ul> |                                                                           |

| WEEK | TOPIC            | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 1-3  | Phases of matter | <p><b>Solids, Liquids and gases</b></p> <p>Briefly define concepts with examples: Solid, Liquid, Gas, Evaporation, Condensation, Freezing, Melting</p> <p>Demonstrate practically the 3 phases in which matter could be found (solids, liquids, gases)</p> <p>Distinguish between solids, liquids and gases with practical examples.</p> <p>Describe the properties of solids, liquids and gases.</p> <p>Describe and explore how matter can change into different phases through heating and cooling using different applicable materials.</p> | <p><b>Note to the teacher:</b> Use as much visual examples as possible.</p> <p>These are only basic definitions to the following concepts:</p> <ul style="list-style-type: none"> <li>• <b>Solid:</b> Solids hold their shape. Solids could be hard, soft, big or small.</li> <li>• <b>Liquid:</b> Liquids flow and take on the shape of the container they are in.</li> <li>• <b>Gas:</b> Gases are everywhere for example the atmosphere are gases that surrounds the Earth.</li> <li>• <b>Evaporation:</b> When a liquid turns into a gas when heat is added.</li> <li>• <b>Condensation:</b> When a gas cools down and forms a liquid.</li> <li>• <b>Freezing:</b> When a liquid forms a solid when cooled.</li> <li>• <b>Melting:</b> When a solid turns into a liquid when heat are added.</li> </ul> <p><b>Step 1:</b></p> <p><i>Demonstrate practically the 3 phases in which matter could be found (solids, liquids, gases).</i></p> <ul style="list-style-type: none"> <li>▪ <i>Introduce the terms solid, liquid and gas. Introduction could also be a <b>practical demonstration</b> by changing ice (solid) into water (liquid) and water into water vapour (gas) or vise versa. Through this demonstration, learners could see that the change was caused by freezing and boiling water which means there was a difference in temperatures.</i></li> </ul> |

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|  |  | <p><b>Step 2:</b></p> <p><i>Distinguish between solids, liquids and gases with practical examples.</i></p> <ul style="list-style-type: none"> <li>▪ <i>Use many examples of materials and ask learners to sort them into 3 groups (solid, liquid, gas) for example wood, stone plastic, fabric, water, juice, tea, air, cooking oil, cooking gas etc. Learners could now describe them.</i></li> </ul> <p><b>Step 3:</b></p> <p><i>Describe the properties of solids, liquids and gases. (There are excellent PowerPoints and videos available on You Tube)</i></p> <p><i>Solids: Particles are packed closely together. Particles cannot move but vibrate. Strong force of attraction keeps the particles together. Has a fixed shape and volume.</i></p> <p><i>Liquids: Small spaces between particles. Particles can move quickly but bump against each other. Force of attraction between particles is weaker than solids. Not easily compressible but takes on the shape of the container.</i></p> <p><i>Gases: Large spaces between particles. Particle can move quickly and easily. Force of attraction between particles is very weak. Gases are easily compressed and take up the shape of its container.</i></p> |
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|  |  | <p><b>Step 4:</b></p> <p><i>Describe and explore how matter can change into different phases through heating and cooling using different applicable materials.</i></p> <ul style="list-style-type: none"> <li>• <b>Demonstrate:</b> <i>Evaporation, condensing, freezing and melting by using water and ice. Take examples of solids like chocolate, ice cream, ice. Ask the learners to predict: “What do you think will happen if we heat these things?”</i></li> <li>• <b>Demonstrate:</b> <i>Melting and solidifying using different substances such as butter, fat, wax, ice cream, chocolate (refer to different workshops at school).</i></li> </ul> <p><b>Examples:</b></p> <ul style="list-style-type: none"> <li>○ <i>Solids turn to liquid when heated (gain energy)</i></li> <li>○ <i>Warm up jelly from the fridge. It melts and becomes a liquid, cool it and it becomes a solid again.</i></li> <li>○ <i>Heat 1 teaspoon of sugar until it melts it becomes a liquid, it solidifies again when cooled (makes a toffee)</i></li> <li>○ <i>Heat any liquid (milk, juice, water or any drink) it boils and turns into water vapour (gas).</i></li> <li>○ <i>Boil a liquid and hold a cold plate above the boiling liquid, droplets of liquid form on the plate (water vapour condenses)</i></li> </ul> |
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|  |  |  | <p><b>Step 5:</b></p> <ul style="list-style-type: none"> <li>▪ <i>Learners write about their observations in each experiment of the changes in phases.</i></li> </ul> <p><b>Step 6:</b></p> <ul style="list-style-type: none"> <li>▪ <i>Learners could identify and describe other examples of phase changes at home (e.g. cooking and baking or when welding in the garage).</i></li> </ul> <p><b>Step 7:</b></p> <ul style="list-style-type: none"> <li>• <i>Assess the learners' knowledge. Can the learner:</i> <ul style="list-style-type: none"> <li>○ <i>Correctly explain how melting, solidifying (freezing), evaporation (boiling) and condensation take place.</i></li> <li>○ <i>Find and describe good examples of the above processes in everyday life or workshops at school.</i></li> </ul> </li> </ul> <p><b>Emphasise safety when working with heat.</b></p> <p><b>Resources:</b></p> <p><i>Solid and liquid materials as used in examples chosen by teacher. A heating and cooling (freezing) resource.</i></p> |
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| 4-6 | Metals and non-metals | <p><b>Properties of metals</b></p> <p>Briefly define concepts:</p> <ul style="list-style-type: none"> <li>Metals; Polymers; Ceramics</li> </ul> <p>Distinguish between metals and non-metals</p> <p>Name different materials in the immediate surroundings and describe their features (e.g. colour, texture etc.)</p> <p>Describe the properties of different classes of metals:</p> <ul style="list-style-type: none"> <li>Appearance; Flexibility; Hardness; Shiny; Strength; Heat conduction</li> </ul> <p>Describe the properties of non-metals</p> <ul style="list-style-type: none"> <li>Dullness; Can break easy; Heat conduction; Ability to rust</li> </ul> <p>Uses of metals and non-metals</p> <ul style="list-style-type: none"> <li>Research and describe how metals and non-metals are used to make products for example furniture made in the workshops at school.</li> </ul> | <p><b>Note to the teacher:</b></p> <p>These are only suggested activities, resources and process notes:</p> <p>These are basic definitions you may use other resources: Make use of visual and practical examples in their surrounding area and workshops at school.</p> <ul style="list-style-type: none"> <li><b>Metals:</b> Are known for conducting heat well. They are strong, shiny and hard. Metals could be shaped without breaking or cracking. Examples: Copper, Aluminium, Zinc, Gold, Silver, Brass, Lead, Iron, Tin, Titanium etc.</li> <li><b>Polymers:</b> Many modern products are polymers. It is a synthetic compound. Examples: Nylon, Polyester, Polystyrene, Teflon, Plastic, Wool, Silk, Rubber etc.</li> <li><b>Ceramics:</b> Manufacturing of any product made from a non-metallic mineral like clay that is exposed to extremely high temperatures. Examples: Porcelain, Tiles, Bricks, Glass, Stoneware etc.</li> </ul> <p><b>Step 1:</b></p> <p>Distinguish between metals and non-metals</p> <ul style="list-style-type: none"> <li>Teacher bring different objects to school and use examples in class for example a book, pencil, bottle, cup, spoon, plate, bowl, wood, iron, steel etc. Classify the objects into any two groups and explain their reasons for placing them in those groups.</li> </ul> <p><b>Step 2:</b></p> <p>Name different materials in their immediate surroundings and describe their features (e.g. colour, texture etc.)</p> <ul style="list-style-type: none"> <li>Discuss the materials made of (paper, wood, metal, glass, ceramic, plastic) referring to the properties of materials: texture, strength, colour, flexibility, hardness and solubility:</li> <li><b>Investigate and compare:</b> Compare and record the properties of some metal objects (such as copper wire, coins, nails, cooking pots, knives and forks) and some non-metal objects (such as a piece of chalk, a stone, a pile</li> </ul> |
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|  |  |  | <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>Assess the learners' knowledge. Can learners: <ul style="list-style-type: none"> <li>Tell the difference between metal, glass, plastic, paper, wood and ceramics ect?</li> <li>Describe the properties of metals and non-metals correctly?</li> </ul> </li> </ul> <p><b>Metals:</b></p> <p>Solid and strong.</p> <p>Could be hammered or bent into different shapes (flexible).</p> <p>Shiny especially when they are new.</p> <p>Cold to touch.</p> <p>Excellent conductors of heat.</p> <p>Has the ability to rust/or become dull.</p> <p><b>Non-metals:</b></p> <p>Could be soft or flexible like rubber.</p> <p>Could be hard and brittle like glass.</p> <p>Do not have a silvery appearance but tend to be dull.</p> <p>Could be grouped into different categories: wood, rubber, plastics, and ceramicsetc.</p> <p>When you touch non-metals they are neither cold nor hot.</p> <p>Not very good conductors of heat.</p> <p>Do not have the ability to rust.</p> |
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|  |  |  | <p><b>Step 4:</b></p> <ul style="list-style-type: none"> <li>• Demonstrate and discuss different ways to make old and dull materials look new/shiny again.</li> </ul> <p>Depending on the material the method will be different.</p> <p>Metals: Polish with a fine cloth/Rubbing with a paste made of pumice powder and water.</p> <p>Non-metals: Clean with a fine cloth/Wash with soap and water.</p> |
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| 7-8 | Processing material | <p><b>Briefly explain concepts:</b></p> <ul style="list-style-type: none"> <li>Combining, Mixing, Setting, Cooking/Heating, Cooling, Drying</li> </ul> <p>Explain that all materials can be processed to make new materials or products.</p> <p>Explain that in Africa people have processed materials e.g. clay pots and bricks etc.</p> <p>Explain that processed materials are useful because of their special properties: strong, durable, waterproof, different colours etc.</p> <p>List different products and list the different raw materials it is made of.</p> <p><b>Practical: Activity</b></p> <p>Demonstrate and explain processes used to change the properties of raw materials:</p> <ul style="list-style-type: none"> <li>Mixing and setting</li> <li>Mixing</li> </ul> | <p><b>Note to the teacher:</b> These are only suggested activities, resources and process notes:</p> <p>These are basic definitions you may use other resources:</p> <ul style="list-style-type: none"> <li><b>Combining:</b> The act of combining is to form a new substance.</li> <li><b>Mixing:</b> When 2 or more substances are added together.</li> <li><b>Setting:</b> Allowing the mixture time to cool down after a certain process.</li> <li><b>Cooking/heating:</b> Increase in temperature.</li> <li><b>Cooling:</b> Decrease in temperature.</li> <li><b>Drying:</b> Allowing the moisture or liquid to be evaporated from the mixture by increased temperatures (sunlight, wind, heating)</li> </ul> <p>Use as much visible examples as possible you may use other examples and resources:</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>Do a practical activity to introduce these concepts; the class could be divided into groups. Each group could get a specific method of processing material. Materials such as plaster, sand, gravel, cement, flour, water, ingredients to make dough, jelly powder, wet clay, etc. could be used.</li> <li>When the learners have finished you could ask the learners: "Is the final product after processing a totally new product?"</li> </ul> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>Show learners examples of processed materials in Africa for example baskets, hats, thatched roofs made from plant fibre such as grasses or reeds, clay pots and bricks.</li> </ul> |
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|                                                                                                        |                                                                           | <ul style="list-style-type: none"><li>• Mixing and cooking</li><li>• Mixing and cooling</li></ul> | <p><b>Step 3:</b></p> <ul style="list-style-type: none"><li>• It is very important that learners understand the importance of processing materials (What are the special properties after processing for example strength, durability, waterproof, different colours etc. depending on example).</li></ul> <p><b>Step 4:</b></p> <ul style="list-style-type: none"><li>• Demonstrate and discuss that the process used could change the properties of raw materials:<ul style="list-style-type: none"><li>○ Mixing and setting (Polyfilla and water makes plaster)</li><li>○ Mixing (Flour and water to make sticky paste/glue)</li><li>○ Mixing and Cooling (Jelly powder and water used to make Jelly)</li><li>○ Mixing and Cooking (Making dough)</li></ul></li></ul> <p>(<b>Hint:</b> Teacher could take photos of the ingredients before the processing took place and the final product after processing)</p> |
| 9 -10                                                                                                  | The weeks allocated for formal assessment are integrated across the term. |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Formal Assessment:                                                                                     |                                                                           |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"><li>• Practical: 40 %</li><li>• Test (Pen and Paper): 60 %</li></ul> |                                                                           |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| WEEK  | TOPIC  | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 1 - 3 | Energy | <p>Energy from the sun</p> <ul style="list-style-type: none"> <li>Explain concepts with examples                             <ul style="list-style-type: none"> <li>Energy (What is energy?)</li> <li>Energy source</li> <li>Food as a source of energy</li> <li>Energy transfer</li> <li>Food chain</li> </ul> </li> <li>Identify the energy source in a food chain (sun) by using a simple food chain.</li> <li>Food serves as a source of energy for plants and animals. Introduce learners to energy transfer between different components in a food chain.</li> <li>Describe the organization of a simple food chain.</li> </ul> | <p><b>Introduction</b></p> <ul style="list-style-type: none"> <li>Ask learners how would they feel during the day if they had no food to eat in the morning? (tired, sleepy, have no energy)</li> <li>Ask learners how they would feel during the day if they had a proper breakfast? (good, have energy)</li> </ul> <p><b>Now learners can make a conclusion:</b></p> <p>YOU NEED FOOD TO HAVE ENERGY FOR DAILY ACTIVITIES!</p> <ul style="list-style-type: none"> <li>What is energy? Energy is the ability to do work.</li> <li>What was the source of energy in this example? The food.</li> </ul> <p><b>Suggested Guideline</b></p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>Show learners an example of a simple food chain. (Make use of resources such as YouTube or posters.</li> </ul> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>Use this example and break it down:                             <ul style="list-style-type: none"> <li>What provided the energy? The sun.</li> <li>What absorbed the energy of the sun? The plants</li> <li>What animal (Herbivore/Omnivore) ate the plants? Depending on example.</li> </ul> </li> </ul> |



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|       |                | <ul style="list-style-type: none"> <li>○ Energy from the sun.</li> <li>○ Producers (plants)</li> <li>○ Users (Herbivore, Carnivores, Omnivores)</li> <li>○ Decomposers</li> </ul>                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>○ What happened next? The animals served as food for other animals.</li> <li>○ Explain to the learner the different users: Herbivore, Carnivores and Omnivores).</li> <li>○ What is the final process of life? Death and bodies are then decomposed by decomposers.</li> </ul> <p><b>Step 3:</b></p> <p>Make use of a variety of different food chains. Learners could even design their own food chains.</p>                                                                                                                                                                                                                                                                                                                                                                |
| 4 - 5 | Energy sources | <p><b>Energy Sources</b></p> <p>Explain the following concepts with examples:</p> <ul style="list-style-type: none"> <li>• Energy sources</li> <li>• Renewable sources of energy e.g. sun, wind and water.</li> <li>• Non-renewable sources of energy e.g. coal, crude oil, natural gas, fuels.</li> </ul> <p>Discuss the classifications of both Renewable- and Non-Renewable sources of energy:</p> | <p><b>Note to the teacher:</b></p> <p>These are only suggested activities, resources and process notes.</p> <p>These are basic definitions of the following concepts:</p> <ul style="list-style-type: none"> <li>• <b>Energy source:</b> Supplies energy.</li> <li>• <b>Renewable:</b> Is generated from natural resources such as sunlight, wind, water and plants(biomass). These sources are renewable because they are quickly replenished and are usually available in a never-ending supply.</li> <li>• <b>Non-renewable:</b> Comes from sources that can't be replenished in a short period of time. We get most of our energy from non-renewable sources which include fossil fuels, oil, natural gas and uranium.</li> </ul> <p><b>Prior knowledge:</b></p> <p>What is energy? The ability to do work.</p> |

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|      |       | <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>Allow learners to name the different sources of energy.</li> </ul> <table border="1"> <tr> <td>Sun</td><td>Coal</td><td>Candles</td></tr> <tr> <td>Wind</td><td>Water</td><td>Paraffin</td></tr> <tr> <td>Fuel</td><td>Gas</td><td>Oil</td></tr> </table> <p>Even secondary sources</p> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>Discuss renewable- and non-renewable sources. There are excellent visual media available on You Tube.</li> <li>Allow the learners to classify the different sources mentioned above as renewable- or non-renewable sources.</li> <li><b>Renewable sources:</b> <p><b>Wind:</b> Wind is the movement of the atmosphere. Wind energy could be used to pump water or to generate electricity. Effective near coastal regions.</p> <p><b>Water:</b> Also known as hydropower. Hydro means water. Water power is one of the oldest known sources of energy in human history. Today it is often used to create electricity using dams.</p> <p><b>Sunlight:</b> Energy from the core of the sun used to convert sunlight energy to electrical energy.</p> </li> </ul> | Sun | Coal | Candles | Wind | Water | Paraffin | Fuel | Gas | Oil |
| Sun  | Coal  | Candles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |         |      |       |          |      |     |     |
| Wind | Water | Paraffin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |         |      |       |          |      |     |     |
| Fuel | Gas   | Oil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |      |         |      |       |          |      |     |     |

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|  |  | <ul style="list-style-type: none"> <li>• <b>Plants (Biomass):</b> Energy from plants. Usually the burning of trees (which) causes air pollution. Modern forms of biomass energy are used for automobile fuel and fueling electric power plants</li> <li>○ <b>Non-renewable sources:</b><br/><br/>Fossil fuels: Sources that formed over 300 million years ago. Fossil fuels are made up of decayed plant and animal matter for example oil, natural gas and coal.<br/><br/><b>Uranium (Nuclear power):</b> Nuclear energy uses the power of the atom to create something very simple: steam power. It generates an incredible amount of power that is used to superheat water, converting it into steam which is then used to create electricity. Nuclear power generates waste that can be dangerous to human health and the environment. The waste takes a long time sometimes thousands of years to become safe.</li> </ul> <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>○ Discuss characteristics of renewable- and non-renewable resources.</li> <li>○ Availability</li> <li>○ Impact on environment.</li> <li>○ How cost efficient is it?</li> </ul> <p><b>Enrichment:</b> Do a poster or make a model</p> |
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| 6 - 7 | Energy transfer           | <p><b>Energy around us</b></p> <p><b>Practical: Activity</b></p> <p>Identify and demonstrate different types of energy transfer</p> <ul style="list-style-type: none"> <li>○ Electrical energy to:</li> <li>○ heat energy</li> <li>○ light energy</li> <li>○ sound energy</li> <li>○ movement energy</li> <li>○ Food energy to movement energy</li> </ul> <p>(Make use of a variety of examples)</p> <p><b>Doing this work practical is very effective.</b></p> | <p><b>Guideline:</b></p> <p>Use as much as visual examples as possible.</p> <p>While learners are busy doing practical work make use of a worksheet so that the learners could write or discuss their findings.</p> <p><b>Examples:</b></p> <ul style="list-style-type: none"> <li>- battery powered radio (transfer is potential energy to sound energy)</li> <li>- electrical fan (transfer is electrical energy to movement energy)</li> <li>- Paraffin lamp (transfer is paraffin to heat energy)</li> <li>- solar light (transfer is sunlight into light energy) etc.</li> </ul> <p><b>Enrichment:</b> Learners could make a model to illustrate any type of energy transfer.</p> |
| 8     | Sound and noise pollution | <p><b>Noise pollution</b></p> <ul style="list-style-type: none"> <li>• Briefly explain the following concepts with examples: <ul style="list-style-type: none"> <li>- Sound; Vibrations; Noise pollution</li> </ul> </li> <li>• Explain what noise pollution is and give practical examples in their environment.</li> </ul>                                                                                                                                    | <p><b>Guideline:</b></p> <ul style="list-style-type: none"> <li>• Take learners to one of the workshops in school where there is a lot of noise pollution or make use of You Tube video clips of airports, factories and construction sites that causes noise pollution etc.</li> <li>• Ask the learners to discuss what they think noise pollution is, and then out of their discussions create a definition for noise pollution before consulting a book. Now explain the concepts mentioned to learners.</li> </ul>                                                                                                                                                                 |

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|  |  | <p>Describe the danger of:</p> <ul style="list-style-type: none"> <li>• Loud sound</li> <li>• Sound to hearing</li> </ul> | <ul style="list-style-type: none"> <li>• These are basic definitions of the following concepts: <ul style="list-style-type: none"> <li>○ <b>Sound:</b> Sound is a vibration or wave that can be heard through a solid, liquid or gas.</li> <li>○ <b>Vibration:</b> Sound comes from vibrations composed of frequencies capable of being detected by ears.</li> <li>○ <b>Noise pollution:</b> Unwanted and annoying degree of sound in a particular area.</li> </ul> </li> </ul> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>○ Explain how sound travels from where it is made to where it is heard through vibrations.</li> </ul> <p><b>Step 2:</b></p> <p><b>Investigation</b></p> <ul style="list-style-type: none"> <li>• Explore making different sounds with different materials. (Tin, Hands, musical instruments, their voices etc.) Loud, Soft, Low and High</li> </ul> <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>• Discuss the dangers of loud sound for people and animals with examples applicable to their surroundings. (Loud noise causes loss of hearing/partial deafness)</li> <li>• Ask learners if they could find solutions for certain situations identified by teacher. For example, what could be done in the workshop to help with noise pollution: Learners could wear ear plugs?</li> </ul> |
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| 9 -10                                                                                                                               | The weeks allocated for formal assessment are integrated across the term. |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"> <li>• Practical: 50 %</li> <li>• Test (Pen and Paper): 50 %</li> </ul> |                                                                           |

| WEEK | TOPIC        | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 1-4  | Planet Earth | <p><b>Features of the Earth</b></p> <p>Explain the following concepts with examples:</p> <ul style="list-style-type: none"> <li>Earth, Habitats, Sun, Continent, Island, Planet, Moon, Star</li> </ul> <p>Describe the features of the Earth</p> <ul style="list-style-type: none"> <li>Shape</li> <li>Size</li> <li>Movement</li> <li>Ability to support life</li> </ul> <p>Describe the different features of habitats on the Earth.</p> <p>Explain the difference between a continent and an island.</p> <p>Identify the islands, oceans and continents on Earth – with their names.</p> <p><b>Practical: Illustrate the Earth through a model or drawings</b></p> | <p><b>Guideline:</b></p> <ul style="list-style-type: none"> <li>It is very effective to use visible resources in your teaching with these topics because it makes it easier for learners to grasp new concepts. For example You Tube , posters, pictures and models.</li> <li>Construct a model of part of the Solar System, Sun, Earth, Moon showing the positions of the three bodies as well as their relative sizes not to scale.</li> </ul> <p><b>Step 1:</b></p> <p>These are only basic definitions to the following concepts:</p> <ul style="list-style-type: none"> <li><b>Earth:</b> The Earth is our home planet. It is the third planet from the Sun. The Earth consist of three parts called the Atmosphere (air), Hydrosphere (water) and the Lithosphere (surface/land). The Earth is the only planet that supports life.</li> <li><b>Habitats:</b> It is an environmental area where living things live for example human, plants, animals or other organisms. There is enough food, shelter, protection and mates for reproduction.</li> <li><b>Sun:</b> The sun is the only star at the centre of our solar system. It is the most important source of energy for life on Earth. The sun is a huge ball of fire shooting light and heat in all directions.</li> <li><b>Continent:</b> Is one of several very large landmasses on Earth. These are Asia, Africa, North America, South America, Antarctica, Europe and Australia.</li> <li><b>Island:</b> An island is a piece of land surrounded by water.</li> </ul> |

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|  |  | <ul style="list-style-type: none"> <li>Describe the Earth as a planet in space.</li> <li>Describe how the Earth, Moon and Sun moves in space.</li> </ul> <p>Discuss the different ways to observe space from the Earth</p> | <ul style="list-style-type: none"> <li><b>Planet:</b> A planet is an astronomical object that is massive enough to be rounded by its own gravity. There are eight planets in our solar system.</li> <li><b>Moon:</b> The moon is the Earth's only natural satellite. A satellite is anything that revolves around another larger object in space. The moon's gravitational influence produces the ocean tides and the slight lengthening of the day. The moon revolves around the Earth. It has holes on it called craters formed by huge rocks from outer space that crashed into it.</li> <li><b>Star:</b> Clouds of dust and gas.</li> </ul> <p><b>Step 2:</b></p> <p>Describe the features of the Earth</p> <ul style="list-style-type: none"> <li>Shape: Spherical shape.</li> <li>Size: 12 756 km in diameter.</li> <li>Movement: The Earth revolves around the Sun in 365 <math>\frac{1}{4}</math> days.</li> <li>Ability to support life: The Atmosphere has oxygen; temperatures are not too hot or too cold. Plants provide food. Two thirds of the surface is covered with water.</li> <li>Discuss where on Earth living organisms would be found. (Habitats – Land, Water, Temperatures)</li> <li>Explain the features of an Island and a continent with pictures and examples.</li> </ul> |
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|  |  |  | <p><b>Step 3</b></p> <ul style="list-style-type: none"> <li>• Discuss the differences between an Island and a Continent</li> </ul> <p>Continents sit on continental <u>lithosphere</u> which is part of <u>plates</u> floating high on Earth's <u>mantle</u>.</p> <p>Islands are either extensions of the *oceanic crust for example volcanic islands or geologically they are part of some continent sitting on continental lithosphere for example <u>Greenland</u>.</p> <p>*<u>Oceanic crust</u> is part of the plates, but it is denser than continental lithosphere, so it floats low on the mantle.</p> <p><b>Guideline:</b></p> <p>The teacher takes the learners outside to observe the sun (safety measurements should be explained and applied)</p> |
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| 5 – 6 | The sun | <p>Describe and discuss the unique features of the Sun:</p> <ul style="list-style-type: none"> <li>• Size, Movement, Distance from the Earth, Temperature, Composition,</li> <li>• How the sun support life on Earth</li> </ul> | <p><b>Guideline:</b></p> <ul style="list-style-type: none"> <li>• It is very effective to use visible resources in your teaching with these topics because it makes it easier for learners to grasp new concepts. For example, You Tube, posters, pictures and models.</li> </ul> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Facts about the Sun:</li> </ul> <p>Size: Diameter at the Sun's equator 1,5 million km (109 times the diameter of the Earth). The mass of the sun is about 333 000 times the mass of the Earth. The volume of the sun is about 1,3 million times the volume of the Earth. (This means you could put about a million Earths inside the moon)</p> <p>Movement: The eight planets revolve around the sun. The Sun is in the centre of our solar system.</p> <p>Distance from the Earth: About 150 million kilometres away from Earth.</p> <p>Temperature: The core of the Sun is 14 million °C.</p> <p>Composition: It is a massive ball of glowing gas. It is made up of hydrogen gas and helium gas. Hundreds of nuclear explosions take place every second and create a temperature of about 14 million °C. This creates enormous amounts of light and heat energy and makes the Sun appear to shine.</p> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>• How the sun supports life on Earth?</li> </ul> <p>There are four conditions needed to support life:</p> <p>Temperature: The Earth's average surface temperature is about 16 °C. Temperatures on the</p> |
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|   |      |                                                                                                                                                                         | <p>Earth's surface range between 71 °C to 90 °C temperatures suited for life.</p> <p>Water: Earth is the only planet that has water in a liquid form. Liquid water covers about 76 % of the Earth's surface.</p> <p>Sunlight: Sunlight provides energy to sustain the food chains on Earth. Plants convert energy from the sun into food. The plants are then eaten and the energy is transferred up the food chain.</p> <p>Oxygen: The Earth is the only planet that has oxygen in its atmosphere. Living things like plants and animals need oxygen to breathe and live.</p>                                                                                        |
| 7 | Moon | <p>Describe and discuss the unique features of the moon:</p> <ul style="list-style-type: none"> <li>Basic facts of the moon (Size, Distance from Earth etc.)</li> </ul> | <p><b>Guideline:</b></p> <p>It is very effective to use visible resources in your teaching with these topics because it makes it easier for learners to grasp new concepts. For example, You Tube, posters, pictures</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>Basic facts about the moon:</li> </ul> <p>Size: The moon is about 3 476 km in diameter. It is airless, waterless and lifeless.</p> <p>Distance from Earth: The moon is about 384 000 km away from Earth.</p> <p>Movement: The moon orbits around the Earth.</p> <p>Surface: The moon has a rocky surface that is covered in craters formed by meteors that crashed into it.</p> |

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| 8 -10                                                                                                                        | The weeks allocated for formal assessment are integrated across the term. |
| Formal Assessment: <ul style="list-style-type: none"> <li>• Practical: 40 %</li> <li>• Test (Pen and Paper): 60 %</li> </ul> |                                                                           |

| WEEK | TOPIC            | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                      | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 1-3  | Digestive system | <p><b>The Digestive System</b></p> <p>Explain concepts with examples:</p> <ul style="list-style-type: none"> <li>• Nutrition</li> <li>• Digestive system</li> <li>• Digestion</li> <li>• Indigestion</li> <li>• Organ of the digestive system</li> </ul> <p>Describe the structure of the digestive system</p> <p>Name the parts of the digestive system</p> <p>Describe the features and functions of the parts of the digestive system</p> | <p><b>Guideline:</b></p> <p>Use as much visible examples as possible. There are excellent resources available on You Tube for example: (<a href="https://www.YouTube.com/watch?v=JnzwbiJuAA">https://www.YouTube.com/watch?v=JnzwbiJuAA</a>) from KidsHealth.</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Explain concepts using examples <ul style="list-style-type: none"> <li>○ <b>Nutrition</b> (Food needed to keep growing organisms healthy)</li> <li>○ <b>Digestive System</b> (System that breaks down the food you eat to make food absorbable into the body)</li> <li>○ <b>Digestion</b> (Allows your body to get the nutrients and energy it needs from the food you eat)</li> <li>○ <b>Indigestion</b> (Another name for an upset stomach, occurs when people eat too much, too fast, or food that don't agree with them. Stress, not enough sleep, smoking, drinking alcohol can make indigestion worse)</li> <li>○ <b>Organs of the digestive system</b> (Mouth→Esophagus→Stomach→Small intestine→Pancreas→Liver→Gallbladder→Large intestine→Rectum→Anus)</li> </ul> </li> </ul> <p><b>Resources:</b> Pictures, Visual media or a model of the digestive system.</p> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>• Describe the structure of the digestive system (make use of visual resources for example a poster of the digestive system, You Tube etc.).</li> </ul> |

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|  |  |  | <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>Name and identify the parts of the digestive system</li> <li><b>Enrichment:</b> Learners could do an activity where they draw the parts of the digestive system very basic.</li> </ul> <p><b>Step 4:</b></p> <ul style="list-style-type: none"> <li>Describe the features and functions of the parts of the digestive system: Mouth: Beginning of digestion, chewing breaks the food into pieces that are more easily digested.</li> </ul> |
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|  |  | <ul style="list-style-type: none"> <li>○ <b>Esophagus:</b> Receives food from your mouth when you swallow. Muscular contractions (peristalsis) delivers food to the stomach.</li> <li>○ <b>Stomach:</b> Container that holds food while enzymes(acids)break food down in a usable form. When content of stomach is sufficiently processed they are released into small intestine.</li> <li>○ <b>Small intestine:</b> 6.7 metres long muscular tube that breaks down food using enzymes released by Pancreas and bile from the liver. Once nutrients have been absorbed and leftover-food residue liquid have pass through the small intestine it then moves to the large intestine.</li> <li>○ <b>Pancreas:</b> Secretes digestive enzymes into the small intestine.</li> <li>○ <b>Liver:</b> Absorb nutrients from small intestine. Bile from the liver is secreted into small intestine and digests fat.</li> <li>○ <b>Gallbladder:</b> Stores and concentrates bile and then releases to help absorb and digest fats.</li> <li>○ <b>Large intestine:</b>1.8 metres long muscular tube thatconnects the small intestine to the rectum. Responsible for processing waste so that emptying of the bowels is easy and convenient. Stool passes through the large intestine by means of peristalsisfirst in a liquid state and ultimately in a solid form.</li> <li>○ <b>Rectum:</b> Connects the large intestine to the anus. Disposing of stool and waste.</li> <li>○ <b>Anus:</b> Last part of the digestive track.</li> </ul> |
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| 4-5 | Nutrients in food | <p><b>Food Groups</b></p> <p>Briefly define concepts</p> <ul style="list-style-type: none"> <li>• Food pyramids</li> <li>• Food groups</li> <li>• Supplements</li> <li>• Fat</li> <li>• Nutrition</li> <li>• Water</li> <li>• Vitamins</li> <li>• Mineral</li> <li>• Carbohydrates</li> <li>• Protein</li> </ul> <p>Categorise food into proteins, carbohydrates, fats, oils, vitamins and minerals.</p> <p>Describe and discuss the functions and importance of various food groups in the body.</p> <p><b>Practical activity:</b></p> | <p><b>Guideline:</b></p> <p>Use as much visible examples as possible. For example, make poster/drawing of the food pyramid.</p> <p><b>Step 1</b></p> <ul style="list-style-type: none"> <li>• Explain concepts with examples from the food pyramid.</li> </ul> <p><b>Step 2</b></p> <ul style="list-style-type: none"> <li>• Introduce learners to the importance of healthy food. <ul style="list-style-type: none"> <li>○ <b>Healthy food</b> is food that will eventually go bad for example fruits, vegetables, dairy products and meats.</li> <li>○ <b>Processed food</b> last longer but is unhealthy for example a bag of chips.</li> </ul> </li> </ul> <p>A healthy diet includes the important nutrients: <b>Carbohydrates</b> (Most important source that provides energy), <b>Proteins</b> (Build-up, maintain and replace tissues in your body), Fats (help brain and nervous system to develop correctly examples of good fats are avocado, tuna, salmon and olive oil), Vitamins (helps you grow and develop like you should), <b>Minerals</b> (Calcium – build bones, <b>Iron</b> – deliver oxygen to blood and lungs), <b>Water</b> (Hydration – controls body temperature through sweating).</p> <p><b>Step 3</b></p> <p><b>Practical activity:</b></p> <ul style="list-style-type: none"> <li>• Learners could look for pictures of the five food groups and categorise them according to the food pyramid</li> </ul> |
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|  |  |  | <p><b>Enrichment activity:</b></p> <ul style="list-style-type: none"> <li>• Learners could look for pictures to make a unhealthy food pyramid.</li> <li>• <b>If possible</b> learners could go to the Consumer Studies class and they could prepare a healthy meal for example a chicken salad which includes all these food groups. While learners are preparing the salad the teacher could introduce them to the functions and importance of the food groups.</li> </ul> <p><b>Or:</b> Discuss the feeding scheme menu at school.</p> |
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| 6 | Nutrition | <p>Explain concepts of a balanced diet</p> <p>Discuss the benefits of a balanced diet</p> <p>Explain and discuss how malnutrition can lead to diseases</p> <p><b>Practical: Learners must draw up a healthy menu for breakfast, lunch and supper for one day</b></p> | <p><b>Introduction:</b></p> <p>Use two different menus or pictures of plates of food (breakfast, lunch, supper) and learners must identify which plates will represent a healthy or unhealthy diet. Learners must explain their answers.</p> <p><b>Step 1:</b></p> <p>What is a balanced diet?</p> <ul style="list-style-type: none"> <li>• A balanced diet contains the <b>correct quantities</b> of nutrients, from the basic food groups, that a person needs each day for healthy growth and activity.</li> <li>• In South Africa, the Provincial Health Restructuring Committee (PHRC) formulated a set of food-based dietary guidelines in May 2003. <ul style="list-style-type: none"> <li>○ Enjoy a variety of foods.</li> <li>○ Be active.</li> <li>○ Make starchy foods the basis of most meals.</li> <li>○ Eat plenty of fresh fruit and vegetables.</li> <li>○ Eat dried beans, peas, lentils and soya regularly.</li> <li>○ You can eat chicken, fish, milk or eggs daily.</li> <li>○ Eat fats sparingly.</li> <li>○ Use salt sparingly.</li> <li>○ Drink lots of clean safe water.</li> <li>○ If you drink alcohol, drink sensibly.</li> </ul> </li> </ul> |
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|  |  |  | <ul style="list-style-type: none"> <li>○ Use foods and drinks containing sugar sparingly and don't have them between meals.</li> </ul> <p><b>Practical Demonstration:</b></p> <ul style="list-style-type: none"> <li>• Learners could plan a healthy menu for one day which will include breakfast, lunch and supper.</li> </ul> <p><b>Hint:</b> Give each learner 3 (three) small disposable plates and allow them to look for pictures of food that will present a balanced diet. Learners could paste these pictures on each plate.</p> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>• Discuss the benefits of a balanced diet: <ul style="list-style-type: none"> <li>○ you have more energy</li> <li>○ maintain weight</li> <li>○ health benefits</li> </ul> </li> </ul> <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>• What is malnutrition and how it can lead to diseases: <ul style="list-style-type: none"> <li>○ When a person does not get enough of certain nutrients.</li> <li>○ Show learners images of children who are malnourished.</li> </ul> </li> </ul> <p>Some diseases caused by malnourishment: Kwashiorkor (a lack of protein), anorexia nervosa (under-nutrition) etc.</p> |
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| 7-8 | Food processing | <p><b>Processing Food</b></p> <p>Explain concepts:</p> <ul style="list-style-type: none"> <li>• Processing</li> <li>• Preserve</li> <li>• Artificial</li> <li>• Preservatives</li> <li>• Supplements</li> </ul> <p>Explain the importance and benefits of preserving food.</p> <p>Explain different methods of food preservation with examples.</p> | <p><b>Introduction:</b></p> <p>Take learners back in history: Why the need to preserve food? How the different cultures used these methods?</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Explain the concepts using practical examples</li> </ul> <p>For example: <b>Processing</b> (Usually refers to foods that are packaged in boxes, cans or bags. It is the transformation of raw ingredients into other forms this involves activities like mincing, liquefaction, or macerating.), <b>Preserve</b> (This involves the preventing of bacteria growth in other words methods to keep the food from going bad. There are many processes to preserve food), <b>Artificial</b> (Processed food often contain additive artificial flavouring and chemical ingredients.), <b>Preservatives</b> (Processed foods have at least one or two preservatives. Preservatives are used to be able to store food for a long period of time, to keep quality and flavour – these preservatives can cause health problems such as allergies or hyperactivity.), <b>Supplements</b> (A supplement is a product intended to add nutritional value to the diet, it is not food this may be found in a powder form, capsules etc. It is generally understood to include vitamins, minerals or fibre.)</p> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>• Explain the importance and benefits of preserving food <ul style="list-style-type: none"> <li>○ Food could be stored for long periods of time</li> <li>○ Food is easy to store</li> <li>○ Food wastage is reduced</li> </ul> </li> </ul> |
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|                                                                                                                              |                                                                           |  | <p><b>Step 3:</b></p> <ul style="list-style-type: none"><li>Explain different methods of food preservation with examples:<br/>Boiling, Drying, Refrigeration, Freezing, Salting, Sugaring, Smoking, Pickling, Canning or bottling, Pasteurization, Vacuum Packing</li></ul> |
| 9 -10                                                                                                                        | The weeks allocated for formal assessment are integrated across the term. |  |                                                                                                                                                                                                                                                                             |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"><li>Practical: 50 %</li><li>Test (Pen and Paper): 50 %</li></ul> |                                                                           |  |                                                                                                                                                                                                                                                                             |

| WEEK | TOPIC    | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                     | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 1-2  | Mixtures | <p>Mixtures of materials</p> <p>Briefly explain concepts:</p> <ul style="list-style-type: none"> <li>• Mixtures</li> <li>• Solutions</li> <li>• Solid</li> <li>• Soluble</li> <li>• Insoluble</li> <li>• Pure Substance</li> </ul> <p>Distinguish between a pure substance and a mixture</p> <p>Briefly explain the properties of a mixture</p> <p>Identify any four methods of separating mixtures into pure substance</p> | <p><b>Guideline:</b></p> <ul style="list-style-type: none"> <li>• Make use of practical demonstrations to explain the following concepts for example:</li> </ul> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Mixture: Mixtures are everywhere you look, most things in nature are mixtures (rocks, ocean, atmosphere, orange juice etc.). A mixture is something that could be separated into two or more substances.</li> <li>• Solution: A solution is when two or more substances dissolves in each other. The substance that dissolves is called the solute. The substance that does not dissolve is called a Solvent. For example, Salt water = water (solvent) + salt (solute).</li> <li>• Soluble: Means the substance is capable of dissolving in a solvent. For example: Sugar is a soluble because it is capable to dissolve in water.</li> <li>• Insoluble: Means the substance is incapable to dissolve in a solvent. For example: Sugar is not soluble in oil.</li> <li>• Pure Substance: A pure substance is made up of only one kind of molecules. It is a substance that cannot be separated by physical means. For example, filtration, evaporation, distillation or chromatography. Examples of pure substances: iron, wood, copper, water, gold, oxygen.</li> </ul> |

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|  |  |  | <p><b>Step 2:</b></p> <p><b>Practical demonstration:</b></p> <ul style="list-style-type: none"> <li>• Distinguish between a substance and a mixture: <ul style="list-style-type: none"> <li>○ A pure substance has only one kind of molecules. Sugar is a pure substance.</li> <li>○ A mixture has a number of pure substances mixed together sugar and cinnamon is a mixture of two materials.</li> </ul> </li> </ul> <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>• Properties of a mixture <ul style="list-style-type: none"> <li>○ The components of a mixture can easily be separated.</li> <li>○ Each component keeps its original properties.</li> <li>○ The proportion of the components is variable.</li> </ul> </li> </ul> <p><b>Step 4:</b></p> <ul style="list-style-type: none"> <li>• Identify four methods to separate mixtures in a pure substance:</li> </ul> <p>These examples are only guideline:</p> <ul style="list-style-type: none"> <li>○ <b>A magnet:</b> (used to separate a metal from non-metal) For example it could be used to separate sand from iron.</li> </ul> |
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|  |  |  | <ul style="list-style-type: none"> <li>○ <b>Filtration:</b> (used to separate a solid from a liquid) The liquid passes through the holes of the filter paper. The solid particles are too big and get stuck. For example: Filtration could be used to separate dirt from some water.</li> <li>○ <b>Evaporation:</b> (used to separate a dissolved solute from a solution) When salty water is warmed the water evaporates leaving behind crystals of salt.</li> <li>○ <b>Distillation:</b> The process of purifying a liquid by boiling a liquid and condensing its vapours. For Example, Salt water is turned into fresh water through distillation. Various forms of fuel, such as gasoline, are separated from crude oil by distillation. Alcoholic beverages are made through distillation. The alcohol is boiled off from the rest of the mixture and collected in a concentrated format.</li> <li>○ <b>Sieving:</b> A sieve is a tool with many holes and that is used to separate smaller particles from larger ones or liquids from solids. A coffee filter could be used to separate the coffee grounds from the water.</li> <li>○ <b>Hand sorting:</b> Physical differences between the materials are easily identified and materials could be sorted by hand. For example: Sorting smarties from jelly tots.</li> </ul> |
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| 3-4 | Solutions as special mixture | <p>Name the ingredients of a mixture.</p> <p>Explain the difference between a mixture and a solution.</p> | <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>The ingredients are solids and/or liquids that are combined by a process to make a new product.</li> </ul> <p><b>Examples:</b></p> <ul style="list-style-type: none"> <li>Air = oxygen + nitrogen + other gases</li> <li>Cement = sand + water + gravel</li> <li>Dirty snow = water + dirt</li> <li>Ocean water = salt + water</li> </ul> <p><b>Step 2:</b></p> <p>A solution is a type of mixture.</p> <ul style="list-style-type: none"> <li>A Solution is when two or more substances dissolves in each other. For example: Salt water.</li> <li>A mixture that is not the same throughout is not a uniform mixture. For example: Sand water.</li> </ul> <p><b>Enrichment:</b> Learners could sort different combinations of substances as a mixture or a solution.</p> |
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| 5-6 | Dissolving | <p>Explain the concept dissolving with examples:</p> <ul style="list-style-type: none"> <li>List what you know about familiar situations when dissolving</li> <li>Investigate the factors of dissolving</li> </ul> <p><b>Practical: Demonstrate</b></p> <ul style="list-style-type: none"> <li>Rate of dissolving</li> </ul> | <p><b>Guideline:</b></p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>Explain the concept dissolving with examples:             <ul style="list-style-type: none"> <li>A solution is made of a substance called solute (for example: sugar) that dissolves in another substance called the solvent (for example: water).</li> <li>Solution→Solute dissolves in Solvent</li> <li>Dissolving is the process when the solute dissolves in a solvent.</li> </ul> </li> </ul> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>Rate of dissolving: The rate of dissolving is the measure of how fast the solute dissolves in a solvent.</li> <li>Investigate the factors of dissolving:             <ul style="list-style-type: none"> <li>Temperature: As the particles in the solvent and solute speed up the process of dissolving. They mix together faster.</li> <li>Agitation: Stirring or shaking the solution also speeds up the process of dissolving.</li> <li>Surface-area: The amount of surface of an object.</li> <li>Grain size of the solute</li> </ul> </li> </ul> |
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|  |  |  | <p><b>Step 4:</b></p> <ul style="list-style-type: none"> <li>• <b>Practical demonstration:</b> Use different examples to investigate the rate of dissolving.</li> </ul> <p>For example: Making a cup of coffee:</p> <ul style="list-style-type: none"> <li>○ How does hot or cold water influence the rate of dissolving?</li> <li>○ How will stirring or shaking the coffee and sugar influence the rate of dissolving opposite to not stirring or shaking? etc.</li> </ul> |
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| 7-8 | Water cycle | <p>The Water Cycle</p> <p>Briefly define concepts:</p> <ul style="list-style-type: none"> <li>• Water cycle</li> <li>• Evaporation</li> <li>• Condensation</li> <li>• Precipitation</li> </ul> <p>Describe states of water on earth</p> <p>Describe how matter change from one phase to another</p> <p>Describe the stages and features of the water cycle</p> <p><b>Practical:</b></p> <p>Draw the water cycle and describe it in your own words</p> | <p><b>Guideline:</b></p> <p>This is only a guideline for the teacher when explaining concepts:</p> <p><b>Step 1:</b></p> <p>Make use of a poster/visual media to explain concepts:</p> <ul style="list-style-type: none"> <li>○ <b>Water cycle:</b> Continues movement of water in a cycle above, below and on the surface of the Earth. The Earth has a limited amount of water. As the water keeps going around and around through evaporation, condensation and precipitation. It is the way water moves between being water vapour (gas) to water (liquid) and back to vapour (gas).</li> <li>○ <b>Evaporation:</b> When the sun heats up water in rivers, lakes and oceans it turns water (liquid) into vapour or steam.</li> <li>○ <b>Condensation:</b> Water vapour in the air gets cold and changes into a liquid forming clouds.</li> <li>○ <b>Precipitation:</b> Occurs when so much water has condensed that the air cannot hold it anymore. The clouds get heavy and water falls back to the Earth in the form of rain, hail, sleet or snow</li> </ul> |
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|  |  |  | <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>• Describe the states of water on Earth: <ul style="list-style-type: none"> <li>○ Solid: Ice/Snow/Hail/Sleet</li> <li>○ Liquid: Rain/Rivers/Oceans/Lakes/Underground water etc.</li> <li>○ Gas: Steam/ Water Vapour</li> </ul> </li> </ul> <p><b>Step 3:</b></p> <ul style="list-style-type: none"> <li>• Describe how matter change from one phase to another:</li> </ul> <p>All matter can change from one state to another under extreme pressure and/or extreme temperature.</p> <p><b>Points of change:</b> Freezing point, Meltingpoint,</p> <ul style="list-style-type: none"> <li>○ Pressure and Boiling point.</li> <li>○ Melting: Solid → Liquid</li> <li>○ Freezing: Liquid → Solid</li> <li>○ Boiling: Liquid → Gas</li> <li>○ Condensation: Gas → Liquid</li> </ul> |
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|                                                                                                                              |                                                                           | <p><b>Step 4:</b></p> <ul style="list-style-type: none"><li>Describe the stages and features of the Water Cycle with the use of visual media:</li></ul> <p>There are four (4) stages in the Water Cycle:</p> <ul style="list-style-type: none"><li><b>Evaporation:</b> When the warmth from the sun causes water from oceans, lakes, streams, ice and soils to rise into the air and turn into water vapour (gas).</li><li><b>Condensation:</b> This is when water vapour in the air cools down and turns into liquid water.</li><li><b>Precipitation:</b> This is when water falls from the clouds in the sky.</li><li><b>Collection:</b> This is when water falls from the clouds as rain, hail, snow or sleet and collects in the oceans, lakes and streams. Most will infiltrate into the ground and will collect as underground water.</li></ul> <p><b>Step 5:</b></p> <ul style="list-style-type: none"><li><b>Practical activity:</b> Describe and draw the Water Cycle in your own words.</li></ul> |
| 9 -10                                                                                                                        | The weeks allocated for formal assessment are integrated across the term. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"><li>Practical: 40 %</li><li>Test (Pen and Paper): 60 %</li></ul> |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| WEEK | TOPIC                  | CONTENT                                                                                                                                                                                                                                                                                                  | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 1-2  | Energy and temperature | <p>Measuring Temperature</p> <p>Explain concepts with examples</p> <ul style="list-style-type: none"> <li>• Heat</li> <li>• Temperature</li> <li>• Celsius scale</li> <li>• Boiling point</li> <li>• Conductor</li> <li>• Insulator</li> <li>• Melting point</li> <li>• Measuring temperature</li> </ul> | <p><b>Note to the teacher:</b></p> <p>Before introducing these concepts, the teacher could consult other resources visual media would be the best resource for learners to grasp these concepts.</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Heat: Heat energy is released when molecules in substances are moving rapidly in a very disorderly way.</li> <li>• Temperature: Is the degree of intensity of hotness or coldness.</li> <li>• Celsius scale: A temperature scale that defines the freezing point of water as 0 degrees and the boiling point of water as 100 degrees.</li> <li>• Boiling point: The temperature at which a liquid begins to boil.</li> <li>• Conductor: A material that allows electricity or heat energy to move through it.</li> <li>• Insulator: A material that allows little or no heat or sound to go into or out of something.</li> <li>• Melting point: The temperature at which something melts.</li> <li>• Measuring temperature: Temperature is measured in degrees Celsius.</li> </ul> |





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| 3-4 | Stored energy in fuels | <p>Fuels</p> <p>Explain concepts with definitions and examples:</p> <ul style="list-style-type: none"> <li>• Fuels</li> <li>• Gas</li> <li>• Candle wax</li> <li>• Fossil fuels</li> <li>• Petrol</li> <li>• Paraffin</li> <li>• Electricity</li> <li>• Safety rules</li> </ul> | <p><b>Note to the teacher:</b></p> <p>If possible, try to take examples of these resources to school to physically show them what it is.<br/>(Safety: do not allow learners to come into contact with these resources without informing them of the dangers or safety rules when handling them or without proper supervision.)</p> <p><b>Step 1:</b></p> <p>These are basic definitions of the concepts. You may use other resources to explain these concepts in detail.</p> <ul style="list-style-type: none"> <li>• Fuels: A substance that can be consumed to produce energy.</li> <li>• Gas: A fossil fuel in its gaseous state.</li> <li>• Candle wax: A solid block of <u>wax</u> with an embedded <u>wick</u> that is burned to provide <u>light</u>.</li> <li>• Fossil fuels: Fossil fuels are buried plant and animal matter that have developed within the Earth over millions of years.</li> <li>• Petrol: A kind of oil that comes from below the ground and that is a source of gasoline and other products.</li> <li>• Electricity: Electricity is the flow of electrical power or charge.</li> </ul> <p>Safety rules: Rules/guidelines to prevent accidents and injuries</p> |
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| 5-6 | Burning fuels and Safety with fire | <p>Burning fuels</p> <p>Investigate, compare and describe various fuels</p> <ul style="list-style-type: none"> <li>• Input energy needed to make them burn</li> <li>• Output energy released such as heat and light</li> </ul> <p>Investigate compare and describe how to keep different fuels burning e.g. candle</p> <p>Safety with fire</p> <p>Discuss safety measurements and safety rules when using various fuels</p> <p><b>Practical activity:</b> Learner should identify (for example 5) safety rules and make a poster to illustrate them.</p> | <p>These are basic definitions of the concepts. You may use other resources to explain these concepts in detail.</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Input energy: Energy put into a system to achieve a result (output).</li> <li>• Output energy: The energy produced by a system.</li> </ul> <p>Input energy needed to make them burn:</p> <p>For example:</p> <p>Input energy needed to provide candle light: oxygen + heat energy(fire) + candle wax.</p> <p>Input energy needed to start a car: Petrol or Diesel.</p> <p>Input energy needed to warm water with a solar system: Heat energy (sunlight).</p> <p><b>Note to the teacher:</b> Make use of other practical examples to explain the concept of input energy needed!</p> <p>Output energy released such as heat and light:</p> <p>For example:</p> <p>Output energy from burning a candle: light energy and heat energy.</p> <p>Output energy from starting a car: Movement energy and Sound energy.</p> <p>Output energy form using a solar system the heat water: Heat energy.</p> |
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|  |  |  | <p><b>Note to the teacher:</b> Make use of other examples to explain the concept of output energy released!</p> <p><b>Step 2:</b></p> <p>Discuss safety measurements and safety rules when using various fuels:</p> <p><b>Note to the teacher:</b> Make use of appropriate safety measurements and rules applicable to the learners' environment. Use visual resources for example poster/You Tube.</p> <p><b>Practical activity:</b> Learner should identify (for example 5) safety rules and make a poster to illustrate them.</p> |
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| 7-8 | Energy and electricity | <p>Cells and battery</p> <p>Briefly define concepts with definitions and examples:</p> <ul style="list-style-type: none"> <li>• Battery(cells)</li> <li>• Electric circuit</li> <li>• Electrical components</li> <li>• Stored energy</li> <li>• Sources of energy</li> </ul> <p>List and explain the different types of batteries and their uses</p> | <p>These are basic definitions of the concepts. You may use other resources to explain these concepts in detail.</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• <b>Battery(cells):</b> A battery is a device that stores electrochemical or electrostatic energy. Batteries come in many shapes and sizes. Some are small as a button. Batteries are used every day by people for many different portable devices, automobiles, boats etc.</li> <li>• <b>Electrical components:</b> Cell/battery, bulb/globe, insulated wires, switch etc.</li> <li>• <b>Stored energy:</b> Sources that has the potential to provide energy.</li> <li>• <b>Sources of energy:</b> There are many different types of energy sources renewable- (wind, water, sunlight, biomass) and non-renewable (fossil fuels, Nuclear) energy.</li> </ul> <p><b>Step 2:</b></p> <p>List and explain the different types of batteries and their uses:</p> <p><b>Note to the teacher:</b></p> <p>These are basic examples. You may use other examples.</p> <p>There are two (2) types of batteries:</p> <ul style="list-style-type: none"> <li>• Disposable/Non-rechargeable batteries called Primary batteries for example: Flash light, TV remote control, Clock etc.</li> <li>• Re-useable/Rechargeable batteries called Secondary batteries for example: Cell phones, digital camera, cordlesspower tool etc.</li> </ul> |
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| 9 -10                                                                                                                               | The weeks allocated for formal assessment are integrated across the term. |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"> <li>• Practical: 50 %</li> <li>• Test (Pen and Paper): 50 %</li> </ul> |                                                                           |

| WEEK | TOPIC                              | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1-5  | Movements of the Earth and planets | <p>Explain concepts and definitions:</p> <ul style="list-style-type: none"> <li>• Solar systems</li> <li>• Star</li> <li>• Earth axis</li> <li>• Rotation</li> <li>• Revolution</li> <li>• Satellite</li> <li>• Reflection of sunlight from the moon</li> <li>• gravity</li> </ul> <p>Explain and demonstrate how the earth moves in relation to the sun around us</p> <p>Day and Night</p> <ul style="list-style-type: none"> <li>• Describe the rotation of the earth on its own axis to explain day and night</li> </ul> | <p>These are basic definitions of the concepts. You may use other resources to explain these concepts in detail. Make use of visual resources for examples pictures, posters or You Tube.</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Solar systems: It is all the planets, the moons, comets, asteroids, dust and gas that orbits around the sun. (Everything in the Solar system orbits around the sun)</li> <li>• Star: Stars are burning gas. Stars are formed in clouds of gas and dust known as the Nebulae. Nuclear reactions at the centre of the star create enough energy to make them shine for at least hundred thousand years.</li> <li>• Earth axis: An axis is an invisible line around which an object rotates or spins. The Earth's axis has a tilt of 23,5°. The tilt causes the seasons.</li> <li>• Rotation: Movement around a central point. The Earth spins around its own internal axis which is tilted 23,5°. One rotation takes 1 day (24hours).</li> <li>• Revolution: When a planet or moon travels around an object it is called a revolution. The Earth revolves around the sun in 365 ¼ days from West to East.</li> <li>• Satellite: A satellite is a moon, planet or machine that orbits a planet or star.</li> <li>• Reflection of sunlight from the moon: Moonlight is the light that reaches Earth from the moon. The light of the moon is actually a reflection of the light from the sun.</li> </ul> |

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|  |  | <p>Moon</p> <p>Explain and draw the different phases of the moon</p> <p>Describe the basic characteristics of the moon</p> <p>Describe ways the moon phases are connected to traditional, cultural activities and beliefs in the society</p> <p><b>Practical Activity:</b> Draw a picture to illustrate the revolution of the Earth around the sun causing the different seasons on Earth.</p> <p>OR: Build a model to illustrate it.</p> | <ul style="list-style-type: none"> <li>Gravity: (A force of attraction that pulls objects together) Earth's gravity is what keeps objects on the ground and causes objects to fall. Gravity is what holds the planets in orbit around the sun and what keeps the moon in orbit around the Earth. The larger an object is the more gravity it has. Because the sun is so large, its powerful gravity attracts all other objects in the solar system towards it.</li> </ul> |
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| 6-7 | Systems looking into space | <p>Rocket systems</p> <p>Explain give definitions of the concepts:</p> <ul style="list-style-type: none"> <li>• Rocket</li> <li>• Telescope</li> </ul> <p>Explain how people travel into space</p> <p>Telescopes</p> <p>Describe how telescopes are used to look into space and gather information</p> <p>Discuss the largest telescope is in South Africa</p> | <p>These are basic definitions of the concepts. You may use other resources to explain these concepts in detail. Make use of visual resources for examples pictures, posters or You Tube.</p> <p><b>Step 1:</b></p> <ul style="list-style-type: none"> <li>• Rocket: Long circular device that is launched into the air. It could also mean a type of engine.</li> <li>• Telescope: A magnifier of distant objects or images.</li> </ul> <p><b>Step 2:</b></p> <p>Explain how people travel in space.</p> <ul style="list-style-type: none"> <li>• Space travel is a dangerous and very expensive. A space shuttle or spacecraft is used to travel in space. In order for a space shuttle to break free of Earth's gravity it has to travel at a speed of about 24 000km/h. A space shuttle will need 1,9 million liters of fuel to launch into space. The combination of high speed, heat and fuel needed for launching can lead to a potentially dangerous situation. Astronauts travel to space. (Excellent video's available on You Tube from NASA)</li> </ul> <p><b>Step 3:</b></p> <p>Describe how telescopes are used to look into space and gather information.</p> <ul style="list-style-type: none"> <li>• There are several types of telescopes that are used to study the universe. The Hubble Space Telescope is currently the largest space telescope. The main purpose of a telescope is to make objects from outer space as bright and large as possible. It allows researchers to expand our knowledge of the universe.</li> </ul> |
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|                                                                                                                                  |                                                                           |  | <p><b>Step 4:</b></p> <p>Discuss the largest telescope is in South Africa.</p> <p>The largest telescope in South Africa is the Southern African Large Telescope (SALT) located close to the town Sutherland in the Karoo. Construction was completed at the end of 2005. Since September 2011 the telescope is releasing its potential as Africa's Giant Eye on the universe.</p> |
| 8 -10                                                                                                                            | The weeks allocated for formal assessment are integrated across the term. |  |                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"><li>• Practical: 40 %</li><li>• Test (Pen and Paper): 60 %</li></ul> |                                                                           |  |                                                                                                                                                                                                                                                                                                                                                                                   |

## Year 4 Term 1

| WEEK | TOPIC        | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                          | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 1-2  | Sense Organs | <p><b>Senses</b></p> <p>Briefly define concepts through demonstration:</p> <ul style="list-style-type: none"> <li>• Senses</li> <li>• Sight(vision)</li> <li>• Smell</li> <li>• Hearing</li> <li>• Sense of touch</li> <li>• Taste</li> </ul> <p>Name and describe animal senses adapted for survival:</p> <ul style="list-style-type: none"> <li>• Use hearing for protection etc.</li> <li>• Use smell to find food</li> </ul> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may use other resources to explain these concepts in detail.</p> <ul style="list-style-type: none"> <li>• Senses: There are 5 senses the body has to react on external stimuli.</li> <li>• Sight(vision): Using your eyes to see.</li> <li>• Smell: Detection is scents or odours using your nose.</li> <li>• Hearing: Detection of sound using your ears.</li> <li>• Sense of touch: Bodily contact with an object.</li> <li>• Taste: The sensation of flavour on one's tongue when in contact with a substance.</li> </ul> <p><b>Step 1:</b></p> <p>The teacher sets up different work stations for learners. They experience stimuli through their different senses:</p> <ul style="list-style-type: none"> <li>• Sight (Kaleidoscope)</li> <li>• Hearing (Tapes, singing, clapping etc.)</li> <li>• Tasting (white substances sugar, salt, maize, flour)</li> <li>• Feeling (different objects in a bag)</li> <li>• Smell (different fruits)</li> <li>• Sensitivity (put one hand in hot water, and then places both in cold water)</li> </ul> |

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|  |  |  | <p>Learners explain and/or write their experience.</p> <p><b>Step 2:</b></p> <ul style="list-style-type: none"> <li>• Learners describe through their experiences or prior knowledge how they think an animal's senses help it to survive in its environment.</li> <li>• Teachers uses pictures of animals (in their environment) to discuss how they use their smell senses to find food.</li> </ul> |
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| 3-5 | Human skeleton | <p><b><u>Human skeleton</u></b></p> <p>Briefly define the basic concepts:</p> <ul style="list-style-type: none"> <li>• Muscles</li> <li>• The skull</li> <li>• The backbone</li> <li>• The ribs</li> <li>• Limbs</li> <li>• Shoulder</li> <li>• Hip girdles</li> <li>• Joints</li> <li>• Pelvis</li> </ul> <p>Demonstrate and explain the number of bones that make up an adult human skeleton</p> <p>Describe the function of the bones of the human skeleton</p> <p>Explain how individual bones in the skeleton are held together</p> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may use other resources (pictures/You Tube /model) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Muscles: Elastic tissue that pull our bones when we move.</li> <li>• The skull: Bone that protects the brain and sense organs.</li> <li>• The backbone: The series of vertebrae extending from the skull to the pelvis.</li> <li>• Ribs: Curved bones of the chest that are joined to the backbone.</li> <li>• Limbs: Arm or leg of a person or animal.</li> <li>• Shoulder: The upper joint of a person's arms and the part of the body between this and the neck.</li> <li>• Hip girdles: Bone structure supporting the lower limbs.</li> <li>• Joints: Where two bones meet.</li> <li>• Pelvis: The large bone frame where the legs are attached.</li> </ul> <p><b>Step 1:</b></p> <p>Demonstrate through visual media the number and function of bones in the human skeleton:</p> <ul style="list-style-type: none"> <li>• Skull:</li> <li>• Backbone:</li> <li>• Ribs</li> <li>• Limbs:</li> </ul> |
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|  |  | <p>Discuss the importance of eating healthy foods to develop healthy and strong bones.</p> <p>Discuss therapy and care for broken bones</p> <p><b>Practical: Draw a poster to illustrate and identify the physical changes in human development</b></p> | <ul style="list-style-type: none"> <li>• Shoulder:</li> <li>• Hips:</li> <li>• Pelvis:</li> </ul> <p><b>Step 2:</b></p> <p>Demonstrate and explain how the different bones are held together in the skeleton:</p> <p>Make use of a model, pictures, You Tube or any other resource.</p> <p><b>Step 3:</b></p> <p>Discuss the importance of eating healthy foods to develop healthy and strong bones:</p> <p>Make use of visual resources.</p> <p><b>Step 4:</b></p> <p>Discuss therapy and care for broken bones:</p> <p>Learners who had broken bones could give feedback on their own experiences. The teacher could explain when a bone is broken and the importance of therapy.</p> |
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| 6-8 | Human Reproductive Reproduction | <p><b><u>Human Reproductive System</u></b></p> <p><b>Reproduction</b></p> <p>Introduce and explain concepts with examples</p> <ul style="list-style-type: none"> <li>• Sex hormones</li> <li>• Egg development</li> <li>• Sperm development</li> <li>• Menstrual cycle</li> <li>• Ovulation</li> <li>• Reproduction</li> <li>• Egg cells</li> <li>• Sperm cells</li> <li>• Reproductive system (males and females)</li> </ul> <p>Explain and describe the prenatal development of a fetus during the three (3) trimesters in the uterus:</p> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/You Tube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Sex hormones:</li> <li>• Egg development:</li> <li>• Sperm development:</li> <li>• Menstrual cycle:</li> <li>• Ovulation:</li> <li>• Reproduction:</li> <li>• Egg cells:</li> <li>• Sperm cells:</li> <li>• Reproduction system (males and females):</li> </ul> <p><b>Step 1:</b></p> <p><b>Note to the teacher:</b></p> <p>There are excellent resources available in books, hospitals, slideshows with real-life pictures, clinics and You Tube etc. A qualified nurse could also come to inform learners about the development of a fetus inside the uterus.</p> |
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|                                                                                                                                  |                                                                           | <p>Explain the physical changes:</p> <ul style="list-style-type: none"><li>• Explain the features in the human organisms development</li></ul> | <p>Explain and describe the prenatal development of a fetus during the three (3) trimesters in the uterus</p> <p><b>Step 2:</b></p> <p>Explain the physical changes:</p> <ul style="list-style-type: none"><li>• Explain the features in the human organisms development:</li></ul> <p>Infant → toddler → Adolescence (teenager) → Adult → Senior citizen.</p> |
| 9 -10                                                                                                                            | The weeks allocated for formal assessment are integrated across the term. |                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"><li>• Practical: 50 %</li><li>• Test (Pen and Paper): 50 %</li></ul> |                                                                           |                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                |

| WEEK | TOPIC           | CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 1-3  | Acids and bases | <p><b><u>Acids and bases</u></b></p> <p>Briefly define concepts:</p> <ul style="list-style-type: none"> <li>• Acids</li> <li>• Bases</li> <li>• Indicators</li> <li>• Universal indicator</li> <li>• Neutral</li> <li>• Neutralise</li> </ul> <p>Group different acids and bases from household substances</p> <ul style="list-style-type: none"> <li>• Describe difference between acids and bases</li> <li>• Explain safe handling methods and disposal of household chemicals</li> </ul> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/You Tube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Acids: A sour corrosive substance.</li> <li>• Bases: A bitter soapy substance.</li> <li>• Indicators: A chemical compound that changes color and structure when exposed to certain conditions it is useful for chemical tests.</li> <li>• Universal indicator: A solution which undergoes several color changes over a wide range of pH's. The color is used to "indicate" pH directly. Universal indicators are usually mixtures of several indicators.</li> <li>• Neutral: Substance with a pH-value of 7.</li> <li>• Neutralise: To cause an acidic solution to become neutral by adding a base to it or to cause a basic solution to become neutral by adding an acid to it</li> </ul> <p><b>Step 1:</b> Explain and discuss the differences between acids and bases from household substances through one or two examples of each identified by the teacher.</p> <p>Examples of possible household substances:</p> <p>(tap water, soaps, shampoos, window-, toilet- and drain cleaners, washing powder, Handy Andy, vinegar, lemon, Brussel sprouts, broccoli, tomato, milk of magnesia etc., cosmetics, anti-acids tablets)</p> |



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|  |  | <p><b>Practical activity</b></p> <ul style="list-style-type: none"> <li>• Test various household substances with blue and red litmus paper and sort them into acids bases and neutrals</li> </ul> | <p><b>Step 2:</b> Test various household substances with blue and red litmus paper and sort them into acids bases and neutrals.</p> <p><b>Practical activity:</b> Learners could work in groups.</p> |
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| 4-6 | Process to purify water | <p><b><u>Process to purify water</u></b></p> <p><b>Clean Water</b></p> <p>Introduce and explain concepts with examples:</p> <ul style="list-style-type: none"> <li>• Natural processes</li> <li>• Pollution</li> <li>• Filtering</li> <li>• Purify</li> <li>• Germs and bacteria</li> </ul> <p>Explain the importance of drinking clean water</p> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/You Tube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Natural processes: Natural processes in nature without human interference.</li> <li>• Pollution: Contamination natural resources.</li> <li>• Filtering: Using a device with holes to remove something unwanted from a liquid or gas.</li> <li>• Purify: To remove impurities.</li> <li>• Germs and bacteria: Tiny microscopic creatures that can make you sick.</li> </ul> <p><b>Step 1:</b></p> <p>Explain the importance of drinking clean water:</p> <ul style="list-style-type: none"> <li>• Suggestion: Make use of a class of clean and dirty water. Teacher ask questions:</li> </ul> <p>Why is it important to drink water? (prior knowledge)</p> <p>How much glasses of water must a person drink each day?</p> <p>Which one of these two glasses will the learner drink? Give a reasons for their answer.</p> |
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|  |  | <p>Demonstrate the different methods of cleaning water e.g. Processes as filtering, boiling, sieving etc.</p> <p><b>Practical Investigation</b></p> <p>Design, draw or make your own water filter</p> | <p><b>Step 2:</b></p> <p>Demonstrate the different methods of cleaning water:</p> <p>Filtering: For example cleaning sandwater. (Resources: Filter, filterpaper, a glass of sandwater, clean glass)</p> <p>Boiling: For example tap water. (Resources: Tap water, kettle or any other heat resource.)</p> <p>Sieving: For example cleaning water containing leaves etc. (Resources: water with leaves, a sieve, bowl).</p> |
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| 7-8                                                                                                                          | Impact of weather on matter and materials                                 | <p><b><u>Impact of weather, temperature and wind on materials.</u></b></p> <p>Impact of weather on materials</p> <ul style="list-style-type: none"><li>Metals</li><li>Non – metals</li></ul> <p>Discuss methods to protect materials against the weather</p> | <p>The teacher should explain difficult concepts that learners might experience. Make use of visual resources.</p> <p><b>Note to the teacher:</b></p> <p>Have a class discussion on the impact the weather, temperature and wind have on materials like wooden and iron objects on the school grounds. Show learners visual media for example pictures of metals and non-metals that have been exposed to these factors. (Findings of learners could be the iron formed rust and the wood got brittle.)</p> <p><b>Step 1:</b></p> <p>Impact of weather on metals and non-metals:</p> <ul style="list-style-type: none"><li>Use physical examples of metals and non-metals so that learners could visually see the impact weather had on these examples.</li></ul> <p><b>Step 2:</b></p> <p>Discuss methods to protect materials against the weather:</p> <ul style="list-style-type: none"><li>Suggestion: Teacher could give learners a variety of products that protect materials against weather. For example, paint, varnish, galvanised etc.</li></ul> <p>Learners should identify which of these products would be used best to protect certain materials</p> |
| 9 -10                                                                                                                        | The weeks allocated for formal assessment are integrated across the term. |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"><li>Practical: 40 %</li><li>Test (Pen and Paper): 60 %</li></ul> |                                                                           |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**Year 4Term 3**

| WEEK | TOPIC             | CONTENT                                                                                                                                                                                                                                                                                                                                                               | Techniques, activities, resources and process notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 1-2  | Electric circuits | <p><b>Electric circuits</b></p> <p>Define concepts</p> <ul style="list-style-type: none"> <li>• Static electricity</li> <li>• Electric circuit</li> <li>• Current electricity</li> <li>• Electric components</li> <li>• Conductor</li> <li>• Insulator</li> </ul> <p>Explain when a circuit is complete.</p> <p>Demonstrate examples of conductors and insulators</p> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/YouTube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Static electricity: Electricity created by friction.</li> <li>• Electric circuit: Pathway that allows an electrical current to pass through.</li> <li>• Current electricity: Electricity that moves from one place to another is called current electricity.</li> <li>• Electric components: (Wire, Switch, Cells, Globe/Bulb, Ammeter, Voltmeter) Components able to conduct electricity.</li> <li>• Conductor: Allows an electrical current to pass through.</li> <li>• Insulator: Not able to conduct an electrical current.</li> </ul> <p><b>Step 1:</b></p> <p>Explain when a circuit is complete:</p> <ul style="list-style-type: none"> <li>• A circuit is complete when the flow of charge is able to move from the positive to the negative of a power source.</li> </ul> |

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| 3-4 | Systems to solve problems | <p><b>Systems to solve problems</b></p> <p>Briefly define the following concepts and compare the strength of the current of their cells:</p> <ul style="list-style-type: none"> <li>• Series</li> <li>• Parallel</li> </ul> <p><b>Practical activity:</b> Make simple electrical systems:</p> <p>Draw, label or build an electric circuit that include all components</p> <p>Explain and draw a diagram how electric circuits in our homes are wired from the power supplied from Eskom</p> | <p><b>Step 2:</b></p> <p>Teacher make use of practical examples of conductors and insulators:</p> <p>Make use of examples from various workshops to illustrate conductors and insulators <b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/YouTube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Series: There is only one path the current can flow.</li> <li>• Parallel: There are more than one path the current can take</li> </ul> <p><b>Step 1:</b></p> <p>Practical demonstration by the teacher: Build a circuit in series and parallel so that learner could observe the difference in strength of the current.</p> <p><b>Step 2:</b></p> <p>Draw, label or build an electric circuit that include all components (wire, switch, cells, globes)</p> <p><b>Step 3:</b></p> <p>Teacher explain or describe how electricity are transported from the power station to each household:</p> |
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|     |                   |                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5-6 | Mains electricity | <p><b>Mains electricity</b></p> <p>Describe concepts with examples:</p> <ul style="list-style-type: none"> <li>• Electric supply</li> <li>• Generator</li> <li>• Turbine</li> </ul> <p>Discuss with examples the different energy sources that are used to produce electricity</p> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/You Tube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Electric supply: How much electricity is available for use and how was the electricity generated? What resource was used to generate electricity?</li> <li>• Generator: System used to create electricity.</li> <li>• Turbine: An engine that has a part with blades that are caused to spin by pressure from water, steam, or air.</li> </ul> <p><b>Step 1:</b></p> <p>Discuss with examples the different energy sources that are used to produce electricity:</p> <ul style="list-style-type: none"> <li>• Hydro-electric power: When water is used to supply electricity. Usually near waterfalls or dams. In South Africa the Gariep dam supplies hydro-electric power to its surroundings.</li> <li>• Solar power: Use of the sunlight to generate electricity. The sun's energy is absorbed by solar panels. These solar panels contain photovoltaic cells (batteries) which convert energy from the sun to electrical energy.</li> <li>• Wind power: When the kinetic energy from the wind are used to generate electricity. Wind power generators are built on wind farms. The first windfarm in South Africa was built near a small town Darling in the Western Cape.</li> <li>• Nuclear power: When Uranium ore is mined and converted to a fuel used at nuclear power plants. Uranium or is not a fossil fuel but also a non-renewable fuel. Nuclear power is made in special power station called nuclear power stations. In South Africa</li> </ul> |
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|  |  |  | <p>there is currently one nuclear power station near Cape Town at Koeberg. Nuclear power has very dangerous radioactive waste material. Radioactive waste can harm living things and even cause death. The waste must be buried deep below the earth in special containers for hundreds of years to become less dangerous.</p> <ul style="list-style-type: none"> <li>• Coal power (Fossil Fuel): Fuels formed by buried plant and animal matter that lived millions of years ago. When it is used up it cannot be replaced because it takes so long to form. Using fossil fuels also causes air pollution to the environment.</li> </ul> |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|     |                                       |                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 7-8 | Saving cost of generating electricity | <p><b>Saving Cost of Generating Electricity</b></p> <p>Briefly explain concepts:</p> <ul style="list-style-type: none"> <li>• Saving</li> <li>• Appliance</li> <li>• Solar energy</li> <li>• Load shedding</li> </ul> <p>Investigate alternative energy sources and how to use it instead of using electricity.</p> <p>Recall better practices in using energy to save cost.</p> | <p><b>Note to the teacher:</b></p> <p>These are basic definitions of the concepts. You may consult other resources (pictures/YouTube) to explain these concepts in detail:</p> <ul style="list-style-type: none"> <li>• Saving: Using less of a resource</li> <li>• Appliance: Device designed to perform a specific task.</li> <li>• Solar energy: Energy from the sun.</li> <li>• Load shedding: Action used to reduce the load on something. Interruption of the electricity supply to avoid excessive load on the generating plant.</li> </ul> <p><b>Step 1:</b></p> <p>Investigate alternative energy sources and how to use it instead of using electricity:</p> <ul style="list-style-type: none"> <li>• Suggested examples for the teacher: (Fire to make food, candle/paraffin lamp to make light, wind/sun to dry washing, solar geyser to warm water, battery power appliances etc.)</li> </ul> <p><b>Step 2:</b></p> <p>Recall better practices in using energy to save cost:</p> <ul style="list-style-type: none"> <li>• Suggested examples for the teacher: (Switch of unnecessary lights/electrical appliances/ Lower the thermostat during the summer/use only enough water in the kettle)</li> </ul> |
|-----|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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| 9 -10                                                                                                                               | The weeks allocated for formal assessment are integrated across the term. |
| <p>Formal Assessment:</p> <ul style="list-style-type: none"> <li>• Practical: 50 %</li> <li>• Test (Pen and Paper): 50 %</li> </ul> |                                                                           |

## Year 4 Term 4

| Week | Topic                                                                                                                                                | Content<br>Revision and Consolidation                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-4  | <ul style="list-style-type: none"> <li>• <b>Life and living</b></li> <li>• <b>Matter and material</b></li> <li>• <b>Energy and change</b></li> </ul> | <ul style="list-style-type: none"> <li>• <b>Sense Organs</b></li> <li>• <b>Human skeleton</b></li> <li>• <b>Human Reproductive System</b></li> <li>• <b>Acids and bases</b></li> <li>• <b>Process to purify water</b></li> <li>• <b>Impact of weather on matter and materials.</b></li> <li>• <b>Electric circuits</b></li> <li>• <b>Systems to solve problems</b></li> <li>• <b>Mains electricity</b></li> <li>• <b>Saving Cost of Generating Electricity</b></li> </ul> |
| 5-10 | External examination                                                                                                                                 | <p>External moderation of school assessment over terms 1, 2 and 3 = 75% of qualification</p> <p>Formal external assessment written test or oral = 25% of qualification</p>                                                                                                                                                                                                                                                                                                |

## **SECTION 4:**

### **ASSESSMENT IN NATURAL SCIENCES**

- 4.1 Introduction**
- 4.2 Assessment Principles**
- 4.3 Managing Assessment**

PUBLIC COMMENT

## Assessment across the three years

The school and the teachers have overall responsibility for the assessment of learners.

Teachers are expected to create a valid, reliable and credible assessment process.

### Year 2 and 3

| Year 2-3              | Formal School-Based Assessments |                               |                               | Final End-of-Year Assessments |
|-----------------------|---------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                       | Term 1                          | Term 2                        | Term 3                        | Term 4                        |
|                       | Practical-50%<br>(25 Marks)     | Practical – 40%<br>(20 Marks) | Practical – 50%<br>(25 marks) | Practical – 40%<br>(20 marks) |
|                       | Test – 50%<br>(25 Marks)        | Test/Exam – 60%<br>(30 Marks) | Test – 50%<br>(25 Marks)      | Test/Exam – 60%<br>(30 Marks) |
| Number of assessments | 2                               | 2                             | 2                             | 2                             |
| Term Report           | 100%                            | 100%                          | 100%                          | 25%                           |
| End of Year           | SBA<br>75%                      |                               |                               | 25%                           |

### Year 4

| Year 4                | Formal School-Based Assessments |                               |                               | Final End-of-Year Assessments |
|-----------------------|---------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                       | Term 1                          | Term 2                        | Term 3                        | Term 4                        |
|                       | Practical-50%<br>(25 marks)     | Practical – 40%<br>(20 marks) | Practical – 50%<br>(25 Marks) | External Exam                 |
|                       | Test – 50%<br>(25 Marks)        | Test/Exam – 60%<br>(30 Marks) | Test – 50%<br>(25 Marks)      |                               |
| Number of assessments | 2                               | 2                             | 2                             | 1                             |
| Term Report           | 100%                            | 100%                          | 100%                          | 100%                          |
| End of Year           | SBA<br>75%                      |                               |                               | 25%                           |

## MARK ALLOCATION

A percentage weighting is prescribed for the tests, practical and examinations for each year.

Teachers can use any total marks for tests, practical and examinations but should be converted to the total weighting percentage for the term as well as year. Practical marks can consist of one (1) or more activities. See minimum mark allocation above.

## CLARIFICATION ON ASSESSMENT PERIODS

### Year 2 and 3:

Term 1 Test assessment to consist of work done in term 1 only

Term 2 Test/Exam assessment to consist of work done in terms 1 and 2

Term 3 Test assessment to consist of work done in term 3 only

Term 4 Exam assessment to consist of work done in terms 3 and 4

### Year 4:

Term 1 Test assessment to consist of work done in term 1 only

Term 2 Test/Exam assessment to consist of work done in terms 1 and 2

Term 3 Test assessment to consist of work done in terms 1, 2 and 3

Term 4 Exam assessment to consist of work done in the year

### Cognitive levels for the assessment of content in Year 2, 3 and 4

|                                                                                                         |                                           |                                                                                       |                                                                             |                                                                             |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Setting tests and tasks for different cognitive levels                                                  | Knowing science and technology            | Understanding science and technology                                                  | Applying scientific and technological knowledge                             | Evaluating, analysing, synthesizing, scientific and technological knowledge |
| Percentages indicating the proportion of low, middle and high order questions in tasks, tests and exams | Low order questions<br><b>70%</b>         | Middle order questions<br><b>20%</b>                                                  |                                                                             | High order questions<br><b>10%</b>                                          |
| Useful verbs to use when setting questions                                                              | State<br>Name<br>Label<br>List and others | Explain<br>Describe<br>Compare<br>Plan<br>Rearrange<br>Give an example and others ... | Predict<br>Compare<br>Design<br>Use knowledge to demonstrate and others ... | Evaluate<br>Suggest a reason<br>Interpret and others ...                    |

#### **4.4 Moderation of Assessment**

#### **4.5 General**

PUBLIC COMMENT

## SECTION 5: RESOURCES

PUBLIC COMMENT